

Dell PowerFlex 4.5.x and PowerFlex Manager 4.6.x

Technical Overview

Notes, cautions, and warnings

 **NOTE:** A NOTE indicates important information that helps you make better use of your product.

 **CAUTION:** A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

 **WARNING:** A WARNING indicates a potential for property damage, personal injury, or death.

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Introduction

This document provides a high-level technical overview of PowerFlex. It describes the key PowerFlex features, the management capabilities of the PowerFlex management platform and information regarding the PowerFlex offerings.

The target audience for this document includes presales, Dell Technologies Support, and customers.

This Dell Technologies product uses an embedded Linux distribution that is licensed to Dell. This embedded Linux is only authorized to be used to operate the Dell product and only the Dell product.

For additional PowerFlex software documentation, go to [PowerFlex software documentation](#).

Common PowerFlex terms and associated acronyms

This section identifies common PowerFlex terms and associated acronyms.

Table 1. Common PowerFlex terms and associated acronyms

Term	Acronym
management virtual machine	MVM
management data store	MDS
PowerFlex management controller	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data server	SDS
storage virtual machine	SVM
storage data replicator	SDR
storage data target	SDT
system configuration reporter	SCR

See the [PowerFlex glossary](#) for more information about PowerFlex terms and acronyms.

Revision history

Table 2. Revisions

Date	Document revision	Description of changes
February 2026	1.5	Technical updates
June 2025	1.4	Terminology updates
May 2025	1.3	Updates for 4.6.2
January 2025	1.2	Updates to PowerFlex management platform resource requirements and physical server requirements
November 2024	1.1	Updates for 4.6.1
May 2024	1.0	Initial release

PowerFlex overview

PowerFlex is an enterprise-class, software-defined block, and file storage solution that is deployed, managed, and supported as a single system. Highly flexible and scalable, PowerFlex allows customers to have a mixture of two-layer (server SAN), single-layer hyperconverged configurations, storage-only, PowerFlex file services, and other mixed architectures within a single deployment.

With PowerFlex, resources such as storage and compute can be scaled together or separately, non-disruptively, and in small increments. The system can scale from a few nodes to hundreds in a cluster, linearly scaling I/O performance and throughput. In addition, PowerFlex supports a broad set of operating environments that include multiple hypervisors, bare-metal operating systems, container management tools, and PowerFlex file services capabilities (NFS, CIFS, and additional file services offerings). Using native asynchronous replication, customers have the flexibility to replicate data across multiple datacenters or to cloud providers such as AWS and Azure. This allows customers to transition application architectures, while supporting multiple generations of applications with disparate architectures in a multi-cloud environment.

PowerFlex offers integrated product options. PowerFlex appliance is a scalable system with flexible form factors that comes pre-configured and validated for fast, easy deployment. PowerFlex rack is a rack-scale system that is manufactured, managed, supported, and sustained as one system for single-end-to-end lifecycle support. PowerFlex custom node is a validated engineering solution that provides nearly all the flexibility of the PowerFlex software offering.

Terminology for management offerings

The following table describes important PowerFlex Manager terms and concepts for the PowerFlex management offerings.

Table 3. Terms and concepts

Term	Description
Bring your own hypervisor	Allows customer to supply a hypervisor to host the Kubernetes VM or environment. Enables setup of the PowerFlex management platform system for management purposes in VMware ESXi or Kubernetes VM.
Co-residency	The PowerFlex storage-only nodes in a protection domain runs the management of the system.
Management data store (MDS)	The PowerFlex management data store is on the dedicated multi-node controller. It is not the same as the data store used in a co-resident option. Kubernetes VMs host the management containers that reside on this data store.
Management virtual machine (MVM)	The PowerFlex Manager UI is hosted in a Kubernetes environment running across a series of Linux machines. The VM version is called the management VM. The operating system is a customized SLES. version maintained by Dell Technologies.
PowerFlex management cluster	The infrastructure underlying the PowerFlex management platform in a dedicated multi-node configuration. It is mandatory in the PowerFlex rack, and optional for PowerFlex appliance. The physical controllers with VMware ESXi hosting PowerFlex management platform can be a three-node or a five-node VMware ESXi cluster with: - PowerFlex as shared storage (PowerFlex management cluster 2.0) - vSan (PowerFlex management cluster 1.0) Use in upgrades from PowerFlex 3.8.x systems only. PowerFlex management cluster also hosts VMs for vCenter, a jump server, and other optional NSX or CloudLink managers.
PowerFlex management platform	The management software stack for PowerFlex Manager runs on physical or virtual Linux instances, which are containers that provide

Table 3. Terms and concepts (continued)

Term	Description
	services. This management solution can be implemented on various infrastructure options.

New features

PowerFlex Manager 4.6.x includes the PowerFlex catalog core package 4.5.4 and above.

This release introduces the following new features:

Table 4. New features

Feature	Details
PowerFlex with Nutanix Cloud Platform	Added support for PowerFlex with Nutanix Cloud Platform.

System requirements and product limits

This section describes the PowerFlex product limits, and the system requirements for components for each consumption model. The product limits summarize the supported capabilities and system limits.

Maximum—and in some cases—minimum values for key system components are dictated by the product limits.

Product limits

The following table summarizes PowerFlex supported capabilities and system limits. PowerFlex Manager is not backward compatible.

NOTE: The supported limits that are described might differ from system-enforced limits in the list output of the `sccli --query_system_limits` command.

Table 5. Supported capabilities and system limits

PowerFlex item	Limits		
	PowerFlex software	PowerFlex appliance	PowerFlex rack
System raw capacity	Up to 16 PB	Up to 16 PB	Up to 16 PB
Maximum raw capacity per SDS	160 TB (medium granularity) 128 TB (fine granularity)	160 TB (medium granularity) 128 TB (fine granularity)	160 TB (medium granularity) 128 TB (fine granularity)
Maximum raw capacity per protection domain	8 PB	8 PB	8 PB
Device size	15.36 TB, maximum on medium granularity storage pool 7.86 TB maximum on fine granularity storage pool	15.36 TB, maximum on medium granularity storage pools 7.86 TB maximum on fine granularity storage pools	15.36 TB, maximum on medium granularity storage pools 7.86 TB maximum on fine granularity storage pools
Maximum SDCs per system	2,048	400 managed nodes per system, including all node types, Unmanaged hosts align to 2,048.	
Maximum SDSs per system	512	400 storage-only or VMware ESXi - based hyperconverged nodes per system or both. The total number of managed nodes (storage-only, VMware ESXi - based hyperconverged, and compute-only) cannot exceed 400.	
Maximum SDSs per protection domain	128	128	128
Maximum fault sets per protection domain	64	64	64
Maximum protection domains per system	256	256	256
Maximum devices per protection domain	8,192	768	768
Maximum devices (drives) per SDS server NOTE: Includes SDPM or NVDIMM devices.	64 On a VMware ESXi server, the maximum devices per SDS are 59.	32	32

Table 5. Supported capabilities and system limits (continued)

PowerFlex item	Limits		
	PowerFlex software	PowerFlex appliance	PowerFlex rack
Maximum acceleration devices of type SDPM or NVDIMM (maximum that is supported in a PowerFlex node)	PowerEdge nodes R640/R740xd/R840: • NVDIMM: 6 PowerEdge nodes: R650/R750 • NVDIMM: 8 PowerEdge nodes: R660/R760 • SDPM: 2	PowerEdge nodes R640/R740xd/R840: • NVDIMM: 6 PowerEdge nodes: R650/R750 • NVDIMM: 8 PowerEdge nodes: R660/R760 • SDPM: 2	PowerEdge nodes R640/R740xd/R840: • NVDIMM: 6 PowerEdge nodes: R650/R750 • NVDIMM: 8 PowerEdge nodes: R660/R760 • SDPM: 2
Maximum storage pools per system	1,024	1,024	1,024
Maximum storage pools per protection domain	64	10	10
Maximum devices per storage pool	300 If more are needed, contact Dell Technologies Support.	300 If more are needed, contact Dell Technologies Support.	300 If more are needed, contact Dell Technologies Support.
Minimum number of SDSs in a storage pool	3	3	4
Minimum storage pool size	720 GB	720 GB	720 GB
Minimum devices (drives) per storage pool	3, one per fault unit	3, one per fault unit	4, one per fault unit
Maximum total number of volumes and snapshots in protection domain	32,768	32,768	32,768
Maximum total number of volumes and snapshots per storage pool	32,768	32,768	32,768
Total size of all volumes per storage pool	4 PB	4 PB	4 PB
Volume size	Minimum: 8 GB, maximum: 1 PB	Minimum: 8 GB, maximum: 1 PB	Minimum: 8 GB, maximum: 1 PB
Maximum file system partitions per volume	15	15	15
Maximum number of snapshots in a single vTree	126	126	126
Maximum total number of volumes and snapshots in the system	131,072 (up to 10,000 volumes and snapshots when deploying NVMe over TCP)	131,072	131,072
Maximum volumes that can be mapped to a single SDC	1,024	1,024	1,024
Maximum volumes that can be mapped using API (scli \REST\WebUI) concurrently	1,024	1,024	1,024

Table 5. Supported capabilities and system limits (continued)

PowerFlex item	Limits		
	PowerFlex software	PowerFlex appliance	PowerFlex rack
Maximum number of volumes to SDC mappings per system	262,143	262,143	262,143
Maximum snapshot policies per system	1,000	1,000	1,000
Maximum number of snapshots a snapshot policy can be defined to retain per vTree (not including locked snapshots)	60	60	60
Maximum data networks in system	8  NOTE: SDCs can connect through a router and exceed this limit.	4  NOTE: SDCs can connect through a router and exceed this limit.	4  NOTE: SDCs can connect through a router and exceed this limit.
Maximum data networks for MDM virtual IP v4 addresses	4	4	4
System over-provisioning factor	5x net capacity per medium granularity layout	5x net capacity per medium granularity layout	5x net capacity per medium granularity layout
Fine granularity maximum compression	10x raw capacity	10x raw capacity	10x raw capacity
RMcache	Minimum: 128 MB, maximum: 300 GB	Not applicable	Not applicable
Maximum user accounts	256	256	256
Maximum number of concurrent logged-in management clients (UI/ REST/CLI)	128	128	128
Maximum number of configured syslog servers	16 (must be an even number) Network configuration for NAS controller must include link aggregation (bonding or teaming) using LACP.	16 (must be an even number) Network configuration for NAS controller must include link aggregation (bonding or teaming) using LACP.	16 (must be an even number) Network configuration for NAS controller must include link aggregation (bonding or teaming) using LACP.
Volumes per local consistency group (snapshot)	1,024	1,024	1,024
SCSI-3 reservation type	Write exclusive registrants only	Write exclusive registrants only	Write exclusive registrants only
Maximum managed PowerFlex nodes per system	Not applicable	400	400
Maximum PowerFlex nodes per resource group	Not applicable	32	32

Table 6. PowerFlex file services limits

PowerFlex item	Limit
Maximum NAS cluster size (number of nodes)	16 (must be an even number) Network configuration for NAS controller must include link aggregation (bonding/teaming) using LACP.

Table 6. PowerFlex file services limits (continued)

PowerFlex item	Limit
Minimum NAS cluster size (number of nodes)	2
Maximum number of NAS clusters per system	1
Maximum file system size	256 TB
Maximum number of file systems	16,384 (1,024*16 nodes)
Maximum number of NAS servers per system	2,048 (128*16 nodes)
Maximum snaps (mounted) per system	46,080
Maximum number of file systems per NAS server	125
Maximum number of file systems plus mounted snaps per NAS server	1,500
Maximum NAS server network interfaces per NAS server	10 + 2 (backup)
Maximum NAS server network interfaces per node	300
Maximum NDMP concurrent sessions per node	20
Maximum NDMP concurrent sessions per cluster	320 (20*16 nodes)
Maximum CIFS servers per system	2,048
Maximum NFS servers per system	2,048
Maximum CIFS shares per node	10,000
Maximum CIFS shares per system	160,000
Maximum NFS exports per node	5,000
Maximum NFS exports per system	80,000
Maximum tree quotas per file system	8,191
Maximum file names per directory	10 million
Maximum sub-directories or files per directory	10 million
Maximum number of home directories	20,000
Maximum CIFS TCP connections	128,000
Maximum NFS TCP connections	128,000
Maximum TCP connections per system	153,600
Maximum unique ACLs per file system	4 million
Maximum CIFS shares + NFS exports per system	160,000 + 80,000
Maximum directories per file system	Greater than 10 billion
Maximum open files/directories	1,024,000
Maximum files per file system	32 billion
Supported NFS versions	v3/v4/v4.1/v4.2
Supported SMB (CIFS) versions	v1/v2/v3/v3.02/v3.1.1
Supported FTP versions	FTP and SFTP

Table 7. NVMe over TCP limits

PowerFlex item	Limit
Maximum volumes mapped to a single NVMe host (Linux)	1024

Table 7. NVMe over TCP limits (continued)

PowerFlex item	Limit
	VMware ESXi 8.0 and VMware ESXi 7.0u3f or higher
Maximum volumes mapped to a single NVMe host (VMware ESXi)	32 with VMware ESXi 7.0u3f 256 using VMware ESXi 8.0
Maximum NVMe hosts connected to system	1024  NOTE: These hosts are in the total number of SDCs per system.
Maximum SDTs per protection domain	128
Minimum SDTs per protection domain	2 Using minimum SDTs may block the ability to reach maximum NVMe hosts.
Maximum SDTs per system	512
Maximum paths in multipathing driver per volume	8 4 in VMware ESXi 7.0u3f
Maximum connections per host per protection domain	16
Maximum NVMe host connections (I/O controllers) per SDT	512
Maximum NVMe host connections (I/O controllers) per system	65,519
Maximum I/O controller queue depth	128
Maximum I/O controller queues	32  NOTE: Number of queues + queue depth is automatically negotiated on connection.
Maximum volume-to-host mappings (SDC/NVMe) per system	262,143

 **NOTE:** Replication is not supported on NVMe over TCP-based volumes.

Table 8. Replication limits

PowerFlex item	Limit
Number of destination systems for replication	4
Maximum SDRs per system	128
Maximum number of replication consistency groups	1,024
Maximum replication pairs in replication consistency group with initial copy	1,024
Maximum number of volume pairs per replication consistency group	1,024
Maximum volume pairs per system	32,000
Maximum number of remote protection domains	6
Maximum number of copies per replication consistency group	1
Maximum ratio of peer SDRs in a source protection domain to a target protection domain	1:6
Maximum ratio of maintenance SDRs in a source protection domain to a target protection domain	1:18
Recovery point objective (RPO)	Minimum: 15 seconds, maximum: 1 hour

Table 8. Replication limits (continued)

PowerFlex item	Limit
Maximum replicated volume size	64 TB

Table 9. PowerFlex Manager limits

PowerFlex management platform capability	Limit
Maximum number of configured users	20
Maximum number of LDAP users	20
Maximum number of LDAP servers	1
Maximum number of DNS servers	1
Maximum number of time sources	1
Maximum number of syslog servers	1
Maximum number of SMTP servers	1
Maximum number of vCenters	1

Table 10. Deployment and NDU limits

PowerFlex item	Limit
Maximum latency for MDM network	200 msec
SDS IP addresses	Must be unique. For example, 127.0.0.1 must not be used, as it refers to several machines.

System requirements

This section lists the requirements for system components for each consumption model.

This section is specific to PowerFlex software deployments.

The following packages are required for each consumption model:

- RCM for PowerFlex rack
- Intelligent Catalog for PowerFlex appliance
- PowerFlex custom node software ZIP
- PowerFlex software ZIP

For more information about software package names and versions, see the *Dell PowerFlex and PowerFlex Manager Release Notes*.

NOTE: PowerFlex rack and PowerFlex appliance software do not contain the VMware ESXi and vCSA images and the patch depot file. Customers must procure and provide the VMware ESXi and vCSA images and patch depot file for building the PowerFlex management cluster, PowerFlex hyperconverged nodes, or PowerFlex compute-only nodes from the Broadcom support portal.

PowerFlex cluster component requirements

Each PowerFlex cluster includes the following components. Before deploying PowerFlex, ensure that the PowerFlex environment meets the server requirements for the cluster type.

Three-node cluster

- One primary MDM
- One secondary MDM

- One tiebreaker
- Minimum of three SDSs (on the same servers as the above components, or on three different servers)
- SDCs, up to the maximum allowed (on the same servers as the above components, or on different servers)

Five-node cluster

- One primary MDM
- Two secondary MDMs
- Two tiebreakers
- Minimum of three SDSs (on the same servers as the above components, or on three different servers) Dell Technologies recommends five SDSs so that the cluster members reside with the SDSs.
- SDCs, up to the maximum allowed (on the same servers as the above components, or on different servers)

PowerFlex management platform options

- PowerFlex rack dedicated management cluster—three-nodes or more
- PowerFlex appliance
 - Option 1: Single node using RAID 5 and VMware ESXi 7.0 operating system
 - Option 2: VMware ESXi cluster with three nodes or five nodes, with vSAN, if there is upgrade (PowerFlex management cluster 1.0) or PowerFlex as shared storage (PowerFlex management cluster 2.0).
- PowerFlex custom node—one node, dedicated PowerFlex management cluster

NOTE: In the one-node configuration, a hypervisor is required, such as VMware ESXi, to host PowerFlex management platform VMs.

- PowerFlex custom node or PowerFlex software: Co-resident PowerFlex management cluster — available for PowerFlex software and custom node offerings.
- PowerFlex software or PowerFlex custom node offerings: Bring your own VMware ESXi environment to host PowerFlex management cluster.

PowerFlex management platform resource requirements

PowerFlex management platform supports the following system requirements for PowerFlex software.

The following table summarizes the requirements for node count, volumes, and SDS devices based on your cluster scale.

Table 11. Cluster scale

System scale	Node count	Volumes	SDS devices in the system
Low scale	From 3 to 10	From 0 to 1,000	From 3 to 240
Medium scale	From 11 to 50	From 1,000 to 10,000	From 241 to 1,200
High scale	From 51 to the maximum limit See Product Limits for the nodes maximum limit.	From 10,000 to the maximum limit See Product Limits for the volume maximum limit.	From 1,201 to the maximum limit See Product Limits for the SDS devices maximum limit.

The following table summarizes the PowerFlex management platform storage and resource requirements for PowerFlex software depending on your cluster scale.

Table 12. PowerFlex software storage and resource requirements

System scale	CPU cores	Storage capacity (GB)	RAM (GB)
Low Scale	8	300	24
Medium Scale	8	350	26
High Scale	14	450	32

The following table summarizes the PowerFlex management platform storage and resource requirements for PowerFlex rack and PowerFlex appliance.

Table 13. PowerFlex rack and PowerFlex appliance storage and resource requirements

CPU cores	Storage capacity (GB)	RAM (GB)
14	450	32

Physical server requirements

The following section lists the physical server requirements for PowerFlex and recommended best practices.

The recommended best practice when installing new nodes is to keep nodes with similar hardware characteristics in their own protection domain. Different CPUs, memory speeds, drive capacity, drive types, and network speeds can affect the overall node performance. PowerFlex Manager does not prevent the mixing of node types within a protection domain. The lowest performing system dictates performance, which might not always be the preferred option.

PowerFlex offers a sizing tool to make the complex work of sizing the physical requirement. Go to the [PowerFlex sizer](#) and review the physical sizing considerations on the page. Dell Technologies recommends using the sizer tool for all and not perform manual sizing.

This table summarizes the physical server requirements for Dell PowerEdge servers (R750/R650).

NOTE: Physical memory requirements for Dell PowerEdge servers (R760/R660) include non-volatile random-access memory (SDPM), Dell PowerEdge servers (R750/R650) include non-volatile random-access memory (NVDIMM).

Table 14. Physical server requirements

Component	Requirement
Processor	One of the following: - Intel or AMD x86 64-bit (recommended) - Intel or AMD x86 32-bit (for Citrix Hypervisor (Xen) only) NOTE: - Only current customers who have installed XenServer or Citrix Hypervisor are supported, new customers are not advised to install XenServer or Citrix Hypervisor. - AMD processors are not supported on VxFlex Ready Node and do not support SDPMs or NVDIMMs. There is no support for fine granularity storage pools on AMD-based servers. CPU threads: - MDM: 5 - SDS: Default 8/maximum 12 - CloudLink: 2 - LIA: 1 - SDR: 8 - NVMe target: 8 - SDC: Default 2/maximum 8 - NVMe initiator: Default 2/max 8 NOTE: The recommended best practice is to have a different non-uniform memory access (NUMA) domain than SDR if it is on the same operating system.
Physical memory	PowerFlex component requirements: RAM (GiB) - SDS FG: $10 \text{ GiB} + ((100 * \text{Number_Of_Drives}) + (550 * \text{Total_Drive_Capacity_in_TiB})) / 1024$ - SDS MG: $5 \text{ GiB} + (210 * \text{Total_Drive_Capacity_in_TiB}) / 1024$ - SDS SDPM or NVDIMM allocation for FG: $\text{SDPM_capacity_in_GiB}$ or $\text{NVDIMM_capacity_in_GiB} = ((100 * \text{Number_Of_Drives}) + (700 * \text{Total_Drive_Capacity_in_TiB})) / 1024$ - MDM: 6.5 GiB - CloudLink: 4 GiB - Operating system: 1 GiB

Table 14. Physical server requirements (continued)

Component	Requirement					
	- LIA: 0.35 GiB - SDC: 6 GiB - NVMe target: 9 GiB - NVMe initiator: 2 GiB - SDR: 8586 MiB + 550 MiB * (max remote PDs count) NOTE: For certain system configurations, the SDS must be configured to be able to use more than one NUMA domain. The SDS is configured by default to have affinity to socket 0 in a server or VM. By default, the SDS is connected only to NUMA 0, and only has access to the memory in NUMA 0. If NUMA 0 (usually half the total memory) is less than the memory required by the SDS, you must allow the SDS access to the memory in the other NUMA. Non-volatile RAM (SDPM): - Node storage capacity less than 41.9 TiB: SDPM size 32 GiB - Node storage capacity 41.9 TiB to 91 TiB: SDPM size 64 GiB - Node storage capacity greater than 91 TiB and less than 112 TiB: SDPM size 96 GiB Or non-volatile RAM (NVDIMM): - Node storage capacity less than 40.75 TiB : NVDIMM size 32 GiB - Node storage capacity 49 TiB to 83.82 TiB: NVDIMM size 64 GiB - Node storage capacity greater than 83.82 TiB and less than 111.76 TiB: NVDIMM size 96 GiB The NVDIMM and NVDIMM battery replacement feature are available in PowerFlex Manager for PowerFlex appliance and PowerFlex rack storage-only nodes. NOTE: PowerFlex does not support using SWAP on storage-only and SVM PowerFlex nodes. The default setting should be the following: - Using SWAP with PowerFlex has the following impact: - Slows the performance of the system - Impacts I/O on the system disks which are used for SWAP; which might cause other functions using the system disk such as to have slow I/Os: - MDM repository-related operations - PowerFlex logs written to disk. See how to How to Permanently Disable Swap Partition in Linux. In addition, the settings for vm.overcommit must be set to the following settings on nodes that have MDM and SDS processes on them: in/etc/sysctl.conf file <pre>vm.overcommit_memory=2</pre> <pre>vm.overcommit_ratio=100</pre> For more information about the implication of this setting, see the Linux overcommit strategies.					
Operating system or boot disk	Disk type	RAID	Endurance	Minimum capacity	Sustained performance IOPS (random 4 KB)	Sustained performance bandwidth (sequential)
	BOSS card	2*M.2 devices in RAID 1	0.5 DWPD over 5 years or 200 PB life span writes	960 GB	50/20K read/write	400/80 [MB/sec] read/write
	NOTE: - Due to a limitation, do not install any of the following on the same operating system disk as the MDM: PowerFlex Manager and PowerFlex Installer. - The customer-provided infrastructure for PowerFlex management (KVM/Linux) disk recommendation is for reference purposes only. Enhance performance by selecting high-performance read/write speed storage solutions such as SSDs or NVMe drives.					

Table 14. Physical server requirements (continued)

Component	Requirement
Disk space	- Disk space formula per node: 1 GB * number of nodes in the system + 1 GB (for log collection) - Required path locations: /opt 1 GB for log collection under /tmp (reused every time a log bundle is created) - For SDC running on Linux or Citrix Hypervisor (XenServer)—1 GiB NOTE: - Only current customers who have installed XenServer or Citrix Hypervisor are supported, new customers are not advised to install XenServer or Citrix Hypervisor. - The minimum operating system size for a three-node cluster is 10 GiB. - For VMware ESXi—640 GiB is the minimum for a boot disk; the PowerFlex Storage VM resides on the boot datastore.
Connectivity	One of the following: 10/25/40/50/100 GbE network Dual-port network interface cards (recommended) Ensure the following: - There is network connectivity between all components. - Network bandwidth and latency between all nodes is acceptable, according to application demands. - Ethernet switch supports the bandwidth between network nodes. - MTU settings are consistent across all servers and switches. - The TCP ports are not used by any other application, and are open in the local firewall of the server. NOTE: You can change the default ports.

NOTE:

- If buying a PowerEdge based solution with a PowerFlex software license, select hardware that is identical to the engineered solution (PowerFlex appliance, PowerFlex rack, PowerFlex custom node).
- PowerEdge configurations that differ from the engineered solution are at risk of encountering issues. Aligning to the engineered solution is critically important when selecting drives on an SDS device.
- Avoid agnostic drives. Select vendor-specific drives to ensure the nodes ship with PowerFlex validated drives.
- To ensure PowerFlex software compatibility, it is recommended to configure a solution using a PowerFlex base (PowerFlex appliance, PowerFlex rack, PowerFlex custom node).

Operating system requirements

The following is a list of operating systems that are supported by PowerFlex.

PowerFlex supports Red Hat Enterprise Linux 9.x (Technology Preview for NVMe over TCP), SLES 15.5 and VMware ESXi 8.0 for NVMe over TCP client. For more information, see the [PowerFlex support matrix](#).

Technology preview features are not fully supported and may not be functionally complete. These features are not suitable for deployment in production. However, these features are provided to the customer as a courtesy and the primary goal is for the features to gain wider exposure with the goal of full support in the future. Customers may find these features useful in a non-production environment. Customers are encouraged to provide feedback and functionality suggestions for a technology preview feature before it becomes fully supported.

Table 15. Operating system requirements

Operating system	Requirement
Linux	Packages required for all components: <i>numactl</i> <i>libaio1</i> <i>wget</i>

Table 15. Operating system requirements (continued)

Operating system	Requirement
	• <i>apr</i> • <i>libapr1</i> • <i>python2-rpm</i> or <i>python3-rpm</i> • <i>yum-utils</i> Additional package required for SDC: • <i>which</i> Additional packages required for MDM components: • <i>bash-completion</i> (for SCLI completion) • Latest version of Python 3.X (for newer operating systems such as CentOS\REDHAT 8.X) • <i>binutils</i> • <i>java-11-openjdk-headless</i> Additional packages needed by SDS: • <i>smartmontools</i> • <i>smartmontoolslibs</i> • <i>binutils</i> • <i>sg3_utils</i> • <i>hdparm</i> • <i>pciutils</i> Additional packages needed by Diag-tools: • <i>sar</i> • <i>sysstat</i> For supported storage controllers, the following packages are also needed: • Dell: <i>perccli</i> • HP raid controllers: <i>hpssacli</i> Additional packages required for SDS components that contain NVDIMMs: • <i>ndctl</i> • <i>jq</i> • <i>daxio</i> • <i>libpmem</i> To use the secure authentication mode, ensure that OpenSSL 64-bit v1.1 is installed on all servers in the system. • <i>libopenssl1_0_0-1.0.1g-0.40.1.x86_64.rpm</i> • <i>openssl1-1.0.1g-0.40.1.x86_64.rpm</i> To use LDAP, ensure that OpenLDAP 2.4 is installed on all servers. Additional packages needed for LIA: <i>pciutils</i> • NVMe initiator: Linux <i>nvme_tcp</i> and <i>nvme_fabrics</i> modules must be loaded in the operating system. Use <code>lsmod grep nvme</code> to validate that they are available.
Windows	• Only SDC and LIA are supported components on Windows. • To install SDC, ensure that Microsoft Security Update KB3033929 is installed. To use the secure authentication mode, ensure that the following are installed on all servers in the system: • OpenSSL 64-bit v1.0.1 or later (v1.1, however, is not supported) • Visual C++ redistributable 2010 package, 64-bit

Table 15. Operating system requirements (continued)

Operating system	Requirement
	The Microsoft Windows Storage Spaces feature (available from Windows 8.1 and later), is not supported in Microsoft Failover cluster running on PowerFlex volumes.
VMware ESXi	Compute-only (SDC) support: VMware ESXi operating systems: 8.0 or 7.0 (SDC only). The latest patches of VMware ESXi and vCenter are tested periodically. For the most updated list, see the PowerFlex support matrix. VMware ESXi has discontinued support for the Intel Xeon Platinum 8100, Gold 6100/5100, Silver 4100, Bronze 3100 (Skylake-SP series) because Intel has transitioned the Skylake generation processors into the End of Servicing Update or End of Servicing Lifetime phase. For more information, see CPU Support Deprecation and Discontinuation in vSphere Releases.

Converting CentOS to SUSE Linux Enterprise Server operating system

PowerFlex Manager automates the process of converting nodes from an embedded operating system based on CentOS to an embedded operating system based on SLES. This feature is available on PowerFlex appliance and PowerFlex rack systems.

In PowerFlex Manager 4.6.x and above, you can convert hyperconverged and storage-only nodes from CentOS to SLES. This feature only applies to PowerFlex rack and PowerFlex appliance deployments on storage-only deployments or SVM nodes that are managed by PowerFlex Manager.

For PowerFlex nodes in lifecycle mode, contact Dell Technologies Support for details about whether a manual procedure is available for your environment.

Storage virtual memory requirements

The following is a list of PowerFlex storage virtual memory requirements.

Storage virtual memory requirements for Dell PowerEdge servers (R660/R760), storage-only nodes include non-volatile random-access memory (SDPM), and storage-only nodes include non-volatile random-access memory (NVDIMM) Dell PowerEdge servers (R650/R750).

Table 16. Storage virtual memory requirements

Component	Memory allocation rules
Memory	PowerFlex RAM (GiB) • SDS FG: 10 GiB + 550 MiB per 1 TiB device size + 100 MiB per device • SDS MG: 5 GiB + 335 MiB per 1TiB device size • MD Read Cache: Total_Drive_Capacity_in_TiB * 8% • MDM: 6.5 GiB • CloudLink: 4 GiB • Operating System: 1 GiB • LIA: 0.35 GiB • SDC: 6 GiB • SDR: 8586 MiB + 550 MiB * (max remote PDs count) NOTE: SDR storage virtual memory support is only available as part of a PowerFlex appliance or the PowerFlex rack managed by PowerFlex Manager. • SDR memory requirement for hyperconverged systems (storage virtual machines): 12 GB NOTE: For certain system configurations, the SDS must be configured to be able to use more than one non-uniform memory access (NUMA) domain. The SDS is configured by default to have affinity to socket 0 in a server or VM, and by default, the SDS is connected only to NUMA 0, and only has access to the memory in NUMA 0. If NUMA 0

Table 16. Storage virtual memory requirements (continued)

Component	Memory allocation rules
	(usually half the total memory) is less than the memory required by the SDS, you must allow the SDS access to the memory in the other NUMA. Non-volatile RAM SDPM: - Node storage capacity less than 41.9 TiB: SDPM size 32 GiB - Node storage capacity 41.9 TiB to 91 TiB: SDPM size 64 GiB - Node storage capacity greater than 91 TiB and less than 112 TiB: SDPM size 96 GiB or non-volatile RAM (NVDIMM): - Node storage capacity less than 49 TiB: NVDIMM size 32 GiB - Node storage capacity 49 TiB to 96 TiB: NVDIMM size 64 GiB - Node storage capacity greater than 96 TiB: NVDIMM size 96 GiB NOTE: PowerFlex does not support using swap on storage-only and storage virtual machine nodes. This should be the default setting. Using swap with PowerFlex has the following impact: - Slows the performance of the system. - Impacts I/O on the system disks which are used for swap. This may cause other functions that use the system disk to have slow I/Os such as: - MDM repository-related operations - PowerFlex logs written to disk Disable swap and set <code>vm.overcommit</code> to the following on nodes that have MDM and SDS processes on them in the <code>/etc/sysctl.conf</code> file: <pre>vm.overcommit_memory=2</pre> <pre>vm.overcommit_ratio=100</pre>
Base storage virtual memory	350 MiB
Tiebreaker MDM	50 MiB

Software defined persistent memory

Software defined persistent memory (SDPM) is supported as non-volatile memory on Dell PowerEdge servers (R660/R760), storage-only and hyperconverged infrastructure.

The features of SDPM include:

- Battery backup provides enough power to store the data in memory.
- A vault optimized storage solution to write or store the data from acceleration memory to a persistent location.
- Compression and high-speed memory performance in shared system memory.

SDPM support for storage-only and hyperconverged deployments on PowerFlex R660 nodes and PowerFlex R760 nodes

This release supports discovery and deployment of PowerFlex storage only and PowerFlex hyperconverged configurations on PowerFlex R660 nodes and PowerFlex R760 nodes. The SDPM nodes can be discovered using RedFish API that is integrated with iDRAC. You can use iDRAC to configure the BIOS and allocate SDPM.

On the **Resources** page for the node, filter by **Physical Disks** to view the details of the vault optimized storage solution under the PCIe Extender section.

Upgrade CloudLink

PowerFlex supports CloudLink version 8.1, a device encryption software.

Upgrade from CloudLink 7.x to CloudLink 8.1.x on existing deployments and install CloudLink 8.1 on new deployments of PowerFlex. For more information about CloudLink documentation and guidelines on how to upgrade from CloudLink 7.x to 8.x, see the *Dell PowerFlex and PowerFlex Manager Administration Guide*.

Browser support

PowerFlex supports Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.

Other requirements

PowerFlex requires that you use a minimum of three SDS servers and three storage devices (a minimum of one disk per SDS), with a combined free capacity of at least 720 GB. These minimum values are true per system and per storage pool.

Architecture

This section presents the main components of the PowerFlex architecture.

System

PowerFlex is based on a configuration of hardware and a software components working together.

Hardware

Hardware can be the existing application servers used by the data center, or a new set of nodes (if, for example, you want to dedicate all nodes solely for running the PowerFlex SAN storage system).

The PowerFlex rack architecture includes a full data center solution including: rack, switches, PowerFlex server nodes, power, and cabling. Built for capacity, performance, or compute per the specific configuration.

- Servers are the basic computer units used to install and run PowerFlex. They can be the same servers used for the applications (server convergence), or a dedicated cluster. In any case, PowerFlex is hardware-agnostic so aside from performance considerations, the type of server is inconsequential.
- The storage media can be any storage media, in terms of the type (SSD or NVMe SSD) and anywhere (external).

Support for Dell PowerEdge R660, R760, R6625, and R7625 servers

PowerFlex certifies support of the following platforms for all offerings.

- Dell PowerEdge R660 and Dell PowerEdge R760 servers (Intel based).
 - Available with a new persistent memory solution, SDPM.
- Dell PowerEdge R6625 and Dell PowerEdge R7625 servers (AMD based) on PowerFlex compute - only nodes.

Software

The PowerFlex virtual SAN consists of the following software components:

- Meta data manager (MDM): Configures and monitors PowerFlex. The MDM can be configured in redundant cluster mode, with three members on three servers or five members on five servers, or in single mode on a single server.

NOTE: It is not recommended to use single mode in production systems, except in temporary situations. The MDMs contain all the metadata that is required for system operation. Single mode has no protection, and exposes the system to a single point of failure.

- Storage data server (SDS): Manages the capacity of a single server and acts as a backend for data access. The SDS is installed on all servers contributing storage devices to PowerFlex. These devices are accessed through the SDS.
- Storage data client (SDC): A lightweight device driver that exposes PowerFlex volumes as block devices to the application that resides on the same server on which the SDC is installed.
- Storage data replication (SDR): A component that acts as a proxy between the SDS and SDC that must be installed for replication.
- NVMe target, also known as storage data target: Manages the translation of I/O from the internal PowerFlex protocol to the NVMe over TCP standard.
- NVMe initiator: NVMe driver that connects to the NVMe over TCP storage system to consume storage, and uses an NVMe multipathing driver for high availability and load balancing.

Depending on the preferred configuration, the software components are installed on the server and create a virtual SAN layer that is exposed to the applications that reside on the servers.

Software defined persistent memory

Software defined persistent memory (SDPM) is supported as non-volatile memory on Dell PowerEdge servers (R660/R760), storage-only and hyperconverged infrastructure.

The features of SDPM include:

- Battery backup provides enough power to store the data in memory.
- A vault optimized storage solution to write or store the data from acceleration memory to a persistent location.
- Compression and high-speed memory performance in shared system memory.

SDPM support for storage-only and hyperconverged deployments on PowerFlex R660 nodes and PowerFlex R760 nodes

This release supports discovery and deployment of PowerFlex storage only and PowerFlex hyperconverged configurations on PowerFlex R660 nodes and PowerFlex R760 nodes. The SDPM nodes can be discovered using RedFish API that is integrated with iDRAC. You can use iDRAC to configure the BIOS and allocate SDPM.

On the **Resources** page for the node, filter by **Physical Disks** to view the details of the vault optimized storage solution under the PCIe Extender section.

MDM cluster

The MDM is the monitoring and configuration agent of PowerFlex and is used mainly for management. A multi-MDM environment consists of one primary MDM, while the others function as secondary or tiebreaker.

The MDM is used for management which consists of migration, rebuilds, and all system-related functions. No I/Os run through the MDM.

To support high availability, three or five instances of MDM run on different servers. In a multi-MDM environment, one MDM is given the primary role, and the others act as secondary or tiebreaker MDMs.

The MDM cluster consists of a combination of primary MDM, secondary MDMs, and tiebreaker MDMs. There is also the standby MDM which is not a part of the cluster.

- **MDM:** An MDM is any server with the MDM package installed and it can be given a manager or a tiebreaker (default) role during installation. This role cannot be changed later without reinstalling the MDM. MDMs have a unique MDM ID, and can be given unique names. Before the MDM can be part of the cluster, it must be promoted to a standby MDM.
- **Standby MDM and tiebreaker:** An MDM and a tiebreaker can be added to a system as a standby. Once added, the standby MDM or tiebreaker is attached, or locked, to that specific system. A standby MDM can be called on to assume the position of a manager MDM or tiebreaker MDM according to how it is installed, when it is promoted to be a cluster member.
- **Manager MDM:** An MDM that can act as a primary or a secondary in the cluster. Manager MDMs have a unique system ID, and can be given unique names. A manager can be a standby or a member of the cluster.
- **Tiebreaker MDM:** An MDM whose sole role is to help determine which MDM is the primary. A tiebreaker can be a standby or a member of the cluster. A tiebreaker MDM is not a manager. In a three-node cluster, there is one tiebreaker; in a five-node cluster, there are two tiebreakers. This ensures that there are always an odd number of MDMs in a cluster, which guarantees that there is always a majority in electing the primary MDM.

The following terms are relevant to the MDM cluster, specifically:

- **Primary MDM:** The MDM in the cluster that controls the SDSs and SDCs. The primary MDM contains and updates the MDM repository, the database that stores the SDS configuration, and how data is distributed between the SDSs in the system. This repository is constantly replicated to the secondary MDMs, so they can take over with no delay. Every MDM cluster has one primary MDM.
- **Secondary MDM:** An MDM in the cluster that is ready to take over the primary MDM role if necessary. In a three-node cluster, there is one secondary MDM thus allowing for a single point of failure. In a five-node cluster, there are two secondary MDMs, thus allowing for two points of failure. This increased resiliency is a major benefit to enabling the five-node cluster.
- **Replica:** An MDM that contains a replica of the MDM repository. This includes the primary MDM and any secondary MDMs in the MDM cluster.

The following table describes the available cluster modes:

Table 17. MDM cluster modes

Cluster mode	Members	Description
Three-node (default)	- Primary MDM - Secondary MDM - Tiebreaker	Three-node cluster has two copies of the repository, thus can withstand one MDM cluster failure.
Five-node	- Primary MDM - Two secondary MDMs - Two tiebreakers	Five-node cluster has three copies of the repository, thus can withstand two MDM cluster failures.
Single-node	Primary MDM	Single-node cluster has only one copy of the repository, thus it cannot withstand failure. It is not recommended to use single-node in production systems.

In addition to the cluster members, you can prepare standby managers and tiebreaker nodes, for a total of thirteen cluster and standby MDMs.

The MDM cluster IP address limit is 16 IP addresses, which include all cluster members (primary, secondary, standby MDM, and standby tiebreaker).

The following figure illustrates a five-node MDM cluster:

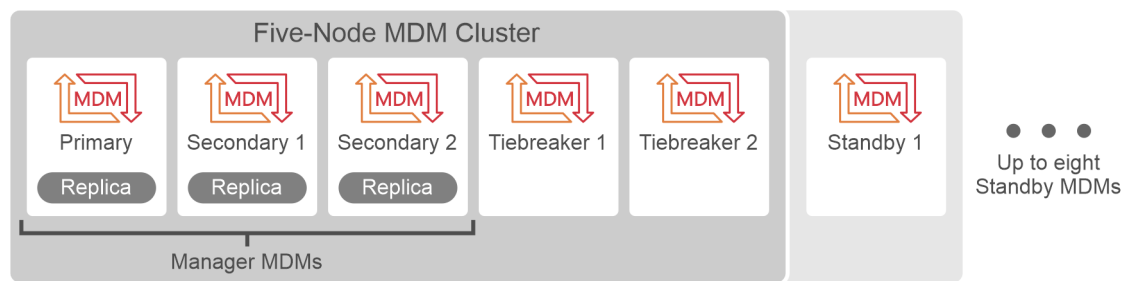


Figure 1. Five-node MDM cluster

All members of the MDM cluster have the same MDM package installed on them.

MDM cluster creation is done automatically when deploying a system with any of the automated deployment tools.

The following list includes best practices for adding MDMs to a multiple-rack system:

- For a large system with multiple racks, the MDM cluster members should be distributed in separate racks for resiliency.
- The MDM network requirements described in the *Networking* section must also apply to the network connections between the racks.
- You can ensure higher resiliency when the rack is shut down for planned or unplanned reasons, especially if fault sets are used. For more information, see *Fault sets*.
- To ensure users do not shut down the MDMs by mistake, mark them physically with stickers saying **Do not shutdown: MDM installed here**, or add a meaningful prefix\suffix to the hostname of the server where MDM is installed. For example: *Rack22U32-MDM* You can also confirm that a different node in the rack is used per server as shown in the table:

Table 18. Nodes in rack

Node level	Rack number	MDM role
Top node	1	MDM
Bottom node	2	MDM
Middle node	3	TB

System stability

The MDM proxy ability augments the network stability capabilities.

In the PowerFlex architecture, each storage data client (SDC) is required to be fully connected to the MDM for control and management of actions. If an SDC is unable to reach the MDM, the SDC moves to disconnected state, which at some point could lead to complete I/O failure.

PowerFlex allows clusters that do not use the MDM virtual IP address capability to use the secondary MDMs as a proxy for all control traffic, in order to provide greater redundancy if a network failure occurs.

If a network failure occurs and the MDM proxy is in use, the management API raises an appropriate alert.

Virtual IP addresses for the MDM cluster

You can define up to four virtual IP addresses for the MDM cluster. SDCs are then mapped to the virtual IP addresses, instead of to static MDM IP addresses. MDMs are sometimes switched during normal operation of the cluster, and the virtual IP address will always be mapped to the active MDM.

The use of virtual IP addresses simplifies maintenance procedures on the MDM cluster, because system components communicate via the virtual IP addresses. Therefore, SDCs do not need to be reconfigured when a server hosting an MDM is replaced.

In new installations in Linux environments, the virtual IP address of the MDM cluster can be added and mapped using the CSV installation file. In VMware environments, virtual IP addresses are mandatory, and configuration is performed using the PowerFlex VMware Installation Wizard, in the Configure SVM stage. The REST API can also be used to add virtual IP addresses to the cluster. In all cases, a virtual IP NIC placeholder must be mapped to each virtual IP address. Ensure that NICs are available for this purpose.

Existing systems may be extended to include additional MDMs in a cluster. The new MDMs should be mapped to the existing virtual IP addresses.

If virtual IP addresses need to be modified, you must use the PowerFlex CLI (SCLI) or the REST API, instead of the CSV installation file or the vSphere plugin, exercising extreme caution. All SDCs then require reconfiguration to reflect the changes made to the MDM cluster. Otherwise, the SDCs cannot communicate with the MDM cluster, and volumes are not accessible.

A maximum of four MDM virtual IP networks are supported.

Storage components

The PowerFlex architecture includes the following storage elements: protection domains, fault sets, and storage pools. Together, these elements link the physical PowerFlex layer to the virtual storage layer.

Protection domains

A protection domain is a logical entity that consists of a group of SDSs that back up each other. Each SDS belongs to one protection domain, so each protection domain is a unique set of SDSs.

The maximum recommended number of nodes in a protection domain is 128. This enables the following:

- Optimal performance
- Reduction of theoretical *mean time between failure* issues
- Ability to sustain multiple failures in different protection domains

You can add protection domains during installation. In addition, you can modify protection domains post-installation with all the management clients.

Fault sets

A fault set is a logical entity that contains a group of SDSs within a protection domain.

In a large cluster that spans across several racks, a fault set can be used as a means to increase the overall cluster resiliency. If one of the racks incurs power loss the use of a fault set prevents a denial of availability. By grouping SDSs in different fault

sets, a verification process occurs so that two copies of data do not reside in the same fault set which increases the cluster resiliency. When the data is written to the SDS device, two copies on two SDS in different fault sets are verified.

When defining fault sets, the term *fault units* refers to either a fault set, or an SDS not associated with a fault set.

There must be enough capacity within at least three fault units to enable mirroring.

If fault sets are defined, you can use any combination of fault units, for example:

- SDS1, SDS2, SDS3
- FS1, SDS1, SDS2
- FS1, FS2, SDS1
- FS1, FS2, FS3

In the following figure, there are three protection domains. The middle protection domain (fully depicted) consists of seven SDSs, each with two storage devices.

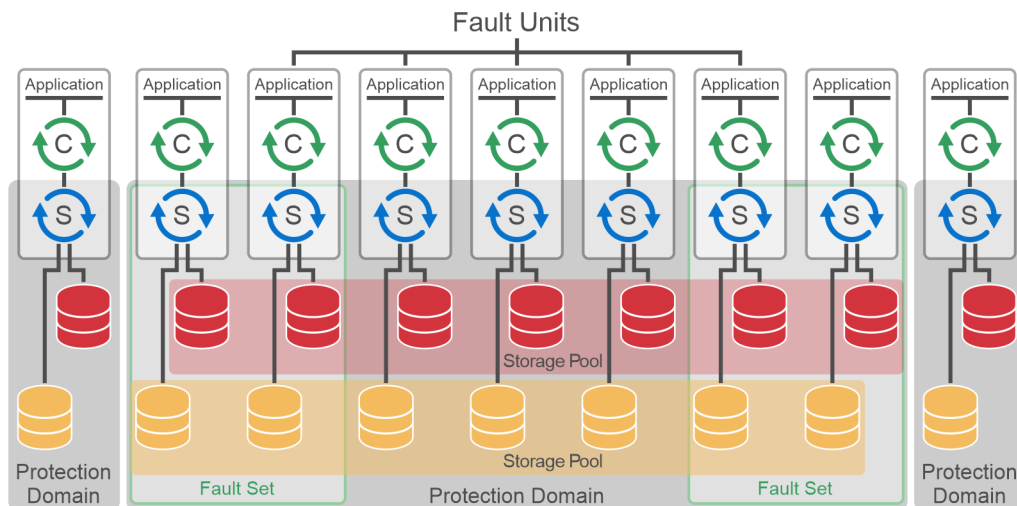


Figure 2. Protection domains, storage pools, and fault sets

Complete the following steps to use fault sets:

1. Ensure that a protection domain exists, or add a new one.
2. Ensure that a storage pool and fault sets (a minimum of three fault units) exist, or add new ones.
3. Add the SDS, designating the protection domain and fault set, and simultaneously, adding the SDS devices into a storage pool.

The automated deployment and installation tools follow this order automatically.

You can only create and configure fault sets before adding SDSs to the system, and configuring them incorrectly may prevent the creation of volumes. An SDS can be added to a fault set only during the creation of the SDS.

You can also add fault sets when adding SDS nodes after initial installation.

Larger PowerFlex storage-only nodes

The maximum SDS is 160 TB (node) on medium granularity storage pools.

- SSD maximum is 15.36 TB (SSD drives)
- Allows higher density systems with larger capacity storage nodes (SDS)
- Initially limited to medium granularity data layout storage pools only (no expansion)

Storage pools

A storage pool is a set of physical storage devices in a protection domain. A volume is distributed over all devices residing in the same storage pool. You can define a magnetic storage pool (for HDDs) or a high-performance storage pool (for SSDs). Storage

pools support medium (1 MB) or fine (4 KB) granularity data layouts and allow enabling or disabling zero padding (the process of zeroing out all the bits of an SDS device).

Storage pools allow the managing of different storage tiers in PowerFlex. Each storage device belongs to one storage pool. The figure shows two storage pools.

When a volume is configured over the virtualization layer, it is distributed over all devices residing in the same storage pool. Each volume block has two copies on two different SDSs. This virtualization layer allows the system to maintain data availability following a single-point failure. If a fault set is configured, then each copy will also be on different fault sets. Two network failures render all the domains in a state where I/O errors are possible on any storage pool.

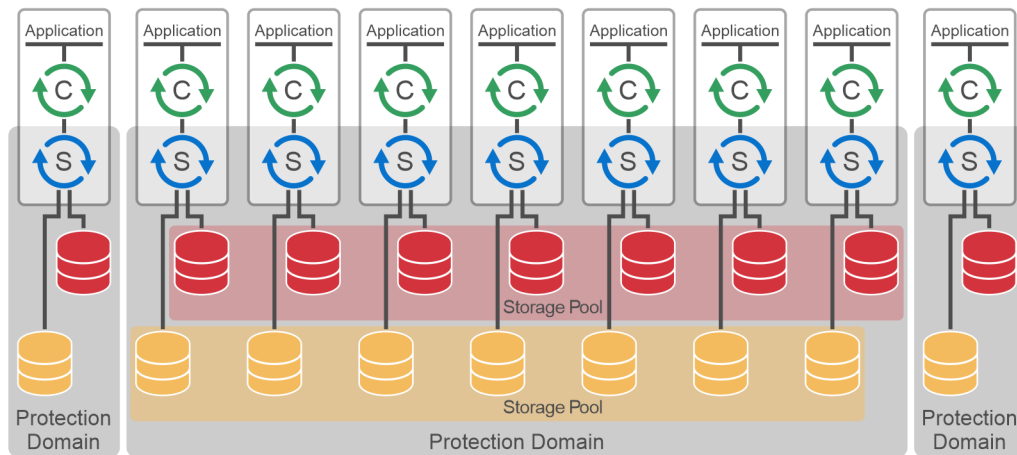


Figure 3. Protection domains and storage pools

You must assign a media type setting to each storage pool. Supported types are: HDD, SSD, and transitional (allows for migration flows).

If all SDSs in a protection domain have two physical drives associated with them, such as one hard drive, and the other SSD, you should define two storage pools:

- Magnetic storage pool
Consists of all HDDs in the protection domain
- High-performance storage pool
Consists of all SSDs used for storage purposes in the protection domain

NOTE:

- Mixing different types of SSDs is not recommended. Creating a separate storage pool for each type is the recommended best practice, for example: SAS SSD, SATA SSD, NVMe SAS SSD.
- PowerFlex might not perform optimally if there are large differences between the sizes of the fault units in the same storage pool. For example, if one device has a much larger capacity than the rest of the devices, performance may be affected. After adding devices, you can define how much of the device capacity is available to PowerFlex by using the SCLI `modify_sds_device_capacity` command.

Storage pools support the following data layouts for HDD or SSD media:

- Medium granularity: Space allocation occurs at 1 MB units.
 - Supports HDD and SSD media
 - Includes persistent checksum for data integrity
- Fine granularity:
 - Requires SSD media and SDPM or NVDIMM for acceleration
 - Space allocation occurs at 4 KB units
 - Includes persistent checksum for data integrity
 - Supports data compression which reduces the size of data that is stored on the disk
 - Supports thin-provisioned, zero-padded volumes

 **NOTE:** Fine granularity and medium granularity storage pools can both exist in a single SDS. You can also migrate volumes across the two layouts.

Each storage pool can work in one of the following modes:

- Zero-padding enabled
Ensures that every read from an area previously not written to returns zeros. Some applications might depend on this behavior. Furthermore, zero padding ensures that reading from a volume will not return information that was previously deleted from the volume.
This behavior incurs some performance overhead on the first write to every area of the volume since the area must be filled with zeros first.
Fine granularity is always zero padded.
- Zero-padding disabled (default only for medium granularity)
A read from an area previously not written to will return unknown content. This content might change on subsequent reads.

Some applications assume that when reading from areas not written to before, the storage will return zeros or consistent data. If you plan to use such applications, then zero padding must be enabled.

You can add storage pools during installation. In addition, you can modify storage pools post installation with most of the management clients.

 **NOTE:** The zero padding policy cannot be changed after the addition of the first device to a specific storage pool.

Acceleration pools

Acceleration pools require non-volatile random-access memory for fine granularity storage pools.

Non-volatile random-access memory acceleration pool features and requirements:

- Must reside on SDPM or NVDIMM devices.
- The SDPM or NVDIMM must be mapped to a DAX device.
- Non-volatile random-access memory acceleration pools cannot be disabled.
- Migration to a different NVRAM pool is not supported.

Limitation: SDPM or NVDIMM failure on SDS node causes failure of all SSDs in the fine granularity storage pool for that node.

An acceleration pool can serve multiple storage pools, but a single storage pool cannot use multiple acceleration pools.

An acceleration device can serve multiple SDS devices, but SDS devices cannot span across multiple acceleration devices.

Data layout

PowerFlex offers a space efficient storage layout with fine granularity.

With fine granularity, since space allocation occurs at 4 KB, the data that is stored on the disk is significantly reduced. Space allocation for medium granularity occurs at 1 MB.

For fine granularity, you must have SDPM or NVDIMM configured as a DAX device. After SDPM or NVDIMM is all set up in your environment, you must configure an SDPM or NVDIMM acceleration pool and then you can select it in the storage pool.

Data-compression is not supported for medium granularity. It supports thin provisioned, zero padded volumes. Zero padding is enabled by default for fine granularity and cannot be disabled. Zero padding is disabled for medium granularity.

Fine granularity supports persistent checksum for data integrity.

Volumes

You must add and configure volumes before enabling applications to access them.

The adding and mapping volume process is necessary, as part of the getting started process, before applications can access the volumes. In addition, you may create additional volumes and map them as part of the maintenance of the virtualization layer.

You can configure the caching option when creating the volumes, or you can change the read RAM cache feature later. If you want to enable the caching feature, ensure that the feature is also enabled in the backend of the system, for the corresponding storage pool and SDSs.

Define volume names according to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query, or via the property sheet for the object in PowerFlex Manager. It is highly recommended to give each volume a meaningful name associated with its operational role.

Naming PowerFlex objects

Provide meaningful names to all PowerFlex objects to make it easier to perform actions such as defining volumes and associating them with applications.

Storage pools can be named *Capacity_Storage* and *Performance_Storage*, which allows you to identify the different tiers.

For protection domains, one example would be separating the SDSs used by the finance department from SDSs used by the engineering department. This separation of different departments is beneficial in many aspects (security being one of them). Thus, you might name the domains *Financial-PD* and *Engineering-PD*.

The fault sets could be called *FS_Rack01* and *FS_Rack02*.

PowerFlex file services

This section describes PowerFlex file system storage and PowerFlex file system support.

File system storage

A file system represents a set of storage resources that provide network file storage.

The storage system establishes a file system that Windows users or Linux/UNIX hosts can connect to and use for file-based storage. Users access a file system through its shares, which draw from the total storage that is allocated to the file system.

Table 19. File system storage components

Component	Description
PowerFlex file services server	A file server that is configured with its network interfaces and other settings exclusively exporting the set of specified file systems through mount points called shares. Client systems connect to a PowerFlex file services server on the storage system to get access to the file system shares. A PowerFlex file services server can have more than one file system, but each file system can only be associated with one PowerFlex file services server.
File system	A manageable container for file-based storage that is associated with the following properties: • A specific quantity of storage. • A particular file access protocol (SMB, NFS, or multiprotocol). • One or more shares (through which network hosts or users can access shared files or folders).
Share or export	A mountable access point to file system storage that network users and hosts can use for file-based storage.
Windows users or Linux/UNIX hosts	A user, host, netgroup, or subnet that has access to the share and can mount or map the resource. For Windows file systems, access to the share is based on share permissions and access control lists that assign privileges to objects defined in Active Directory. For Linux/UNIX file systems, access is permitted based on NFS access settings.

Shares and exports

Shares represent mount points through which users or hosts can access file system resources. Each share is associated with a single file system and inherits the file system protocol (SMB or NFS) established for that file system. Shares of a multiprotocol file system can be either SMB or NFS.

Access to shares is determined depending on the type of file system:

- Windows (SMB) shares: Access is controlled by SMB share permissions and the ACLs on the shared directories and files. For example, you can configure share permissions using the Microsoft Computer Management utility.
 - Active Directory SMB servers: Configure access for users and groups using Windows directory access controls. User/group authentication is performed through Active Directory.
 - Stand-alone SMB servers: Manage a stand-alone SMB server within a workgroup from a Microsoft Windows host.
- Linux/UNIX (NFS) exports: Hosts access is defined by the NFS access control settings of the NFS export. Use PowerFlex to configure access for individual Linux/UNIX hosts or IP address subnets.

All shares within a single file system draw from the total quantity of storage allocated for the file system. Consequently, storage space for shares is managed at the file system level.

PowerFlex file services servers

PowerFlex file services servers provide access to file systems. Each PowerFlex file services server supports Windows (SMB) file systems, Linux/UNIX (NFS) exports, or both. To provide isolated access to a file system, you can configure a file server to function as independent file server with server-specific DNS, NIS, and other settings. The IP address of the PowerFlex file services server provides part of the mount point that users and hosts use to map to the PowerFlex file services storage resource, with the share name providing the rest. Each PowerFlex file services server exposes its own set of file systems through the file system share, either SMB or NFS.

Once a PowerFlex file services server is running, you can create and manage file systems and shares on that PowerFlex file services server.

NOTE: You can create a file system only if there is a PowerFlex file services server running on the storage system. The types of file systems that you can create are determined by the file sharing protocols (SMB, NFS, or multiprotocol) enabled for the file services server.

PowerFlex file services server physical requirements

PowerFlex file nodes can be configured as small, medium, or large, according to the following specifications for Dell PowerEdge R650.

Table 20. Dell PowerEdge R650

Configuration	Cores	RAM (GB)	NICs	Local storage (GB)
Small	2 * 12 (24)	128	4 * 25 GbE	BOSS-S2 480 GB * 2 M.2
Medium	2 * 16 (32)	256	4 * 25 GbE	BOSS-S2 480 GB * 2 M.2
Large	2 * 28 (56)	256	4 * 25/100 GbE	BOSS-S2 480 GB * 2 M.2

CPU

- 2 * Intel® Xeon® Gold 5317 2.8G, 12C, 150W
- 2 * Intel® Xeon® Gold 6346 3.1G, 16C, 205W
- 2 * Intel® Xeon® Gold 6348 2.6G, 28C, 235W

Memory

- 128 GiB: 16 * 8 GiB RDIMM, 3200 MT/s, single rank
- 256 GiB: 16 * 16 GiB RDIMM, 3200 MT/s, dual rank

PowerFlex file nodes can be configured as small, medium, or large, according to the following specifications for Dell PowerEdge R660.

Table 21. Dell PowerEdge R660

Configuration	Cores	RAM (GB)	NICs	Local storage (GB)
Small	2 * 12 (24)	128	4 * 25 GbE	BOSS-S2 480 GB * 2 M.2
Medium	2 * 16 (32)	256	4 * 25 GbE	BOSS-S2 480 GB * 2 M.2
Large	2 * 28 (64)	384	4 * 25/100 GbE	BOSS-S2 480 GB * 2 M.2

CPU

- 2 * Intel® Xeon® Silver 4410Y 2G, 12C/24T, 16 GT/s, 30M Cache, Turbo, HT (150W) DDR5-4000
- 2 * Intel® Xeon® Gold 6426Y 2.5G, 16C/32T, 16 GT/s, 38M Cache, Turbo, HT (185W) DDR-4800
- 2 * Intel® Xeon® Gold 6448Y 2.1G, 32C/64T, 16 GT/s, 60M Cache, Turbo, HT (225W) DDR5-4800

Memory

- 128 GB: 16 * 8 GB RDIMM, 4800 MT/s, single rank
- 256 GB: 16 * 16 GB RDIMM, 4800 MT/s, dual rank
- 384 GB: 24 * 16 GB RDIMM, 4800 MT/s, dual rank

Network

The requirements are similar when customers use bring-your-own hardware.

PowerFlex file services support

This section describes PowerFlex file services support.

- NAS Server and file system cloning
 - PowerFlex file users can clone their NAS server and file system for environment repurposing.
 - Clone a user file system and an entire NAS server.
 - Select any existing NAS server or file system and clone it.
 - Clone a file system from an existing snapshot.
- File-level retention (FLR)
 - File-level retention is a feature for protecting files from modification or deletion until a specified retention period. Protecting a file system using file-level retention enables you to create a permanent, and unalterable set of files and directories. File-level retention ensures data integrity and accessibility, simplifies archiving procedures for administrators, and improves storage management flexibility.

There are two levels of file-level retention:

- Enterprise: Protects data from changes that are made by users and storage administrators using SMB, NFS, and FTP. An administrator can delete a file-level retention enterprise file system which contains locked files.
- Compliance: Protects data from changes that are made by users and storage administrators using SMB, NFS, and FTP. An administrator cannot delete a file-level retention compliance file system which contains locked files. File-level retention compliance complies with the SEC rule 17a-4(f).
- Multi-tenancy
 - NAS servers can be used to enforce multi-tenancy. This is useful when hosting multiple tenants on a single system, such as for service providers. Since each NAS server has its own independent configuration, it can be tailored to the requirements of each tenant without impacting the other NAS servers on the same appliance.
 - Each NAS server is logically separated from each other, and clients that have access to one NAS server do not inherently have access to the file systems on the other NAS servers. File systems are assigned to a NAS server upon creation and cannot be moved between NAS servers.
- Support for PowerFlex file services deployments for PowerFlex rack and PowerFlex appliance:
 - Ability to deploy new PowerFlex file services resource groups and import existing PowerFlex file services resource groups for PowerFlex rack and PowerFlex appliance. This enables the capabilities of the **File** menu.
 - Deployment of compute-only resource groups for PowerFlex file services.
 - PowerFlex file services template settings.
 - PowerFlex file sample template, which can be cloned to generate a template that has the correct settings.

PowerFlex Manager configures the required network on the PowerFlex node and access server-facing ports, and the required network on the control plane. When you deploy a PowerFlex file services template, PowerFlex Manager automatically creates control volumes for the deployment.

- Support for new PowerFlex file services deployments for software management :
 - Ability to deploy new PowerFlex file services resource groups for software management environments. PowerFlex Manager does not support importing existing deployments for software management PowerFlex file services compute-only environments.
 - PowerFlex file services template settings, and support for resource group operations such as node expansion, node removal, and maintenance mode.
 - **PowerFlex-File - SW Only** sample template can be cloned to generate a template that has the correct settings.

 **NOTE:** Installation, network configuration, and setup of the host operating system are prerequisites that the system administrator must perform before running the PowerFlex file services deployment from PowerFlex Manager.

- Support for PowerFlex file services operations for PowerFlex rack and PowerFlex appliance:
 - Ability to expand the number of nodes, remove nodes from the resource group, and put nodes in maintenance mode. If necessary, you can also remove the resource group as a whole.
 - Ability to expand the cluster up to a maximum of 16 nodes. PowerFlex Manager requires a minimum of two nodes in a PowerFlex file services cluster.
 - Ability to remove PowerFlex file services nodes from the resource group, but not actually delete them. When you remove a resource, it PowerFlex Manager removes the deployment information, but does not make configuration changes to the resource itself. You can remove a PowerFlex file services resource group, which only removes the resource group from the appliance. The cluster still exists and can be added back to the appliance.
- Support for PowerFlex file services serviceability:
 - Embedded service enabled integration for PowerFlex file services alerts.
 - SNMP support for PowerFlex file services.
 - Backup.
 - Troubleshooting bundle.
 - CloudIQ integration.
- Single namespace:
 - Ability to create a global namespace supported by the PowerFlex file services cluster with a single export.
 - Allows all hosts with correct access permission to access existing and newly added file systems to the namespace without explicitly mounting them on each client.
 - Consists of several file systems that may be SMB or NFS.
- Common Event Publishing Agent:
 - Support for Common Event Publishing Agent, which is part of the Common Event Enabler package.
 - Ability to receive file event notifications.
 - Ability to use Common Event Publishing Agent to see events on some or all my NFS and SMB file systems.

Replication

PowerFlex enables native asynchronous replication for PowerFlex storage-only and PowerFlex hyperconverged configurations.

Replication can be used to quickly and easily recover from a physical or logical disaster, to migrate data, to test data at a remote site, or to offload backup. The PowerFlex implementation is designed to allow a sub-minute recovery point objective reducing the data-loss to minimal in case of disaster recovery. Replication is elastic and can scale online by adding or removing nodes, flexible, and easy to manage. It enables instant test and failover operations.

Replication peer systems

Replication occurs between two PowerFlex systems, connected by WAN, and designated as peer systems.

PowerFlex supports up to four peer systems from a single source.

 **NOTE:** A source volume can be replicated to just a single target.

Storage data replicator

Storage data replicators process all I/Os of replication volumes.

The source storage data replicator processes all application I/Os of replicated volumes. At the source, the SDC sends application I/Os to the storage data replicator. The I/Os are sent to the target storage data replicators and stored in their journals. The journals of the target storage data replicators apply the I/Os to the target volumes. A minimum of two storage data replicators are deployed at both the source and target systems to maintain high availability. If one storage data replicator fails, the MDM directs the SDC to send the I/Os to an available storage data replicator.

The source storage data replicator processes all application I/Os of replicated volumes. The source storage data replicator saves changes to the journal, and later sends them to the target storage data replicators. At the target, the storage data replicators receive the changes, and when a consistent image is available, apply the changes to the target volumes.

A minimum of two SDRs are deployed at the source and target protection domain to ensure high availability.

The ratio of SDRs in a source protection domain to the storage data replicators in its target protection domain is limited. The maximum ratio of peer storage data replicators is 1:6, and the maximum ratio of storage data replicators in maintenance is 1:18. The Add_SDR and Remove_SDR commands fail if the operation results in an unsupported ratio. If the storage data replicators ratio exceeds the allowed value as a result of a failure or planned maintenance, the Add_RCG command fails. Existing replication consistency groups and replication operations are not impacted.

Journal

Storage data replicators use the journal to hold the replication data.

The source storage data replicators accumulate the data changes in the journal until sending them to the target. The target storage data replicators accumulate received data in the target journal until a completely consistent image is received and can be applied to the target volumes.

SDR journal capacity

To assign enough storage capacity for the replication journal consider the following factors when allocating journal capacity.

- The storage pools from which to allocate the journal capacity. The journal is shared between all the replicated replication consistency groups in the protection domain. Journal capacity should be allocated from storage pools as fast as (or faster than) the storage pool of the fastest replicated application in the protection domain. It should use the same drive technology and about the same drive count and distribution in nodes.
- The minimal requirements needed, for which the default minimum journal size is 400 GB. The minimum journal size can also be calculated as 28 GB per SDR session (where the number of SDR sessions is the number of SDRs + 1) or the capacity needed to sustain an outage, whichever is greater.
- Expected outage time.
 - The minimal outage allowance is one hour, but at least three hours are recommended when computing the minimum size of the replication journal.
 - Journal capacity needed per application is: **maximal application throughput × maximum outage interval**.
- Journal capacity is calculated as a percentage of storage pool capacity, based on the previously calculated needs. Journal capacity should be at least 5% of replicated usable capacity in the protection domain, including volumes used as source and targets.

For example:

- An application generates 1 GB/s of writes.
- The maximal supported outage is 3 hours (3 hours × 3600 seconds = 10800 seconds).
- The journal capacity needed for this application is 1 GB/s × 10800 s = ~10.547 TB.
- Since the journal capacity is expressed as a percentage of the storage pool capacity, divide the 10.547 TB by the size of the storage pool, which is 200 TB: $100 \times 10.547\text{TB} / 200\text{TB} = 5.27\%$ Round this up to 6%.
- Repeat this for each application being replicated.

When a protection domain has several storage pools and several replicated applications, the journal capacity should be calculated as in the example above, and the capacity can be divided among all the storage pools (provided they are fast enough). For higher availability, the journal capacity should be allocated from multiple storage pools.

 **NOTE:** When the storage pool capacity is critical, capacity cannot be allocated for new volumes or for expanding existing volumes. This behavior must be taken into account when planning the capacity available for journal usage. The volume

usage must leave enough capacity available in the storage pool to allow provisioning of journal volumes. The plan should account for the storage pool staying below critical capacity even when the journal capacity is almost fully used.

It is important to note that since journal capacity is defined as a percentage of the total storage capacity in the storage pool, increasing the total storage capacity by adding devices will increase the journal capacity. Similarly, if you decrease the total storage capacity by removing devices from the storage pool, the journal capacity will automatically decrease.

Replication consistency group

The replication consistency group is a logical container for volumes whose application data must be replicated consistently to each other.

For example, if you replicate a database and its transaction log, you need the remote copy of each of the database volumes to reflect an image from the same point in time. You can achieve this by placing all the database volumes in one replication consistency group.

Replication is always defined in the scope of a protection domain. All the objects that participate in the replication are contained in the protection domain, including the volumes in a replication consistency group. The journal capacity from the storage pool in a protection domain is shared among all the replication consistency groups in the protection domain. The SDRs in the protection domain manage I/Os directed to replicated volumes within the protection domain.

The following replication attributes are among those that are defined in the replication consistency group:

- Recovery point objective — The maximum data loss (in units of time) that you are willing to lose. When you set the recovery point objective, this is the goal of replication.
- Direction — Local to remote or remote to local. When there is no replication, for example, following failover or switchover, the direction attribute indicates the direction of the last replication.
- Abstract state — State of replication, for which the options are as follows:
 - **OK** — Replication is healthy, and the recovery point objective target is met.
 - **Recovery point objective violation** — Replication is healthy, but the recovery point objective target is not met.
 - **Error** — Replication is not healthy. The **Error** state attribute provides additional information about the cause for the error.
 - **Stopped by user** — Replication is not carried out due to user action, such as pause or terminate.
- Error state — Describes the error that is preventing data replication.
- Pause mode — When the replication consistency group is paused, all application I/Os are stored in the source journal. When replication of the replication consistency group is resumed, the source SDR sends the journal contents to the target SDR to be applied to the target volumes. You may want to pause an replication consistency group to handle a network issue between the peer systems or when fixing a hardware issue.
- Activity state — When the replication consistency group is active, replication is enabled. When the replication consistency group is inactive, replication is stopped, but replication consistency group configuration is maintained. The replication consistency group can be automatically inactivated by the system (due to insufficient journal capacity), or can be terminated manually by the user. When activating an inactive replication consistency group, the replication consistency group starts replication from the beginning with a full initialization.
- Failover type — The failover operation requested by the user. For more information, see [Accessing target volumes](#).
- Failover state — The states are transient, except for None (no failover) or Done (the failover operation has completed).

Replication pairs

A replication pair is a pair of volumes with one volume at the source system and one at the target system. The data from the source volume is replicated to the target volume.

The replication consistency group can contain several replication pairs. A volume can be a member of at most one replication pair. When the replication pair is created, both the source volume and the target volume must be the same size. It is recommended, but not mandatory, that the volumes in the replication pair have the same attributes (including zero padding and granularity), not doing so can impact performance and capacity.

Recovery point objective

The recovery point objective is the maximum data loss, which is measured in units of time that would be lost in a disaster or outage.

PowerFlex uses journal-based replication to achieve very low recovery point objective and ensure minimal data loss. To ensure recovery point objective compliance, PowerFlex replicates at least twice for every recovery point objective period. For example, setting the recovery point objective to one minute means that PowerFlex can immediately return to operation at the target system with loss of only one minute of data. In order to achieve a recovery point objective of one minute, replication takes place at least every 30 seconds. Recovery point objective compliance is calculated according to when the application data arrives at the target journal. Achieving the recovery point objective is the most important consideration in replication management. The user-defined recovery point objective is the goal of replication. Even so, recovery point objective compliance is not guaranteed. The system reports recovery point objective compliance for each replication consistency group.

Data protection enables replication to multiple systems from one source system to up to four target systems.

Replication I/O flow

For replicated volumes, application data is processed by SDRs at the source and passed to the SDRs at the target.

Application I/Os (both reads and writes) intended for replication volumes are sent from an SDC to an SDR. The source SDR packages the data and distills it so that only the most recent writes are included. The source SDR sends the data over the WAN to the target SDR. At the target system, the SDR processes the replicated data and applies it to the volumes.

Replication auto-provision

PowerFlex supports automatic creation and mapping of a volume in the peer system.

For a given volume on the source system, you can:

- Create a volume on the destination system.
- Create a replication pair for the given source volume and the new destination volume.

In addition, you can request to map the destination volume to an SDC.

Replication initial copy

When replication is first activated for a replication consistency group, the target volumes must be synchronized with the source volumes.

For each replication pair, the entire contents of each source volume are copied to the corresponding target volume when replication is active. This occurs when the replication consistency group containing the replication pair is activated, or when the replication pair is added to an already active replication consistency group. When there is more than one replication pair in the replication consistency group, the order in which the volumes are synchronized is determined by the order in which the replication pairs were activated. You can manually override this order, and also pause and resume the initial copy process, using CLI commands.

The initial synchronization is carried out while the applications are running and performing I/O. Any writes to an area of the volume that has already been synchronized will be sent to the journal. Writes to an area of the volume that has not already been synchronized will be ignored, as the updated content will be copied over eventually as part of the synchronization.

The system limits the number of volumes that undergo **Initial Copy** at any point in time. New volume pairs added to existing replication consistency groups are given priority over new replication consistency groups. This means that replication pairs in a newly activated replication consistency group may not start **Initial Copy** immediately.

Activating or terminating a replication consistency group

When a replication consistency group is active, replication is enabled. When a replication consistency group is inactive, replication is stopped and changes are not tracked, but the replication consistency group maintains its configuration.

By default, a new replication consistency group is created as active. You may, however, specify that it be created in the inactive state.

When the replication consistency group is in an inactive state, replication is stopped and the journal capacity is released. The I/Os are handled by the SDS and not the SDR and no journal capacity is consumed. All replication consistency group configurations are maintained, including source and target roles, pair configuration, and volume access modes.

The replication consistency group can be terminated automatically by the system when the journal capacity is fully consumed. You can also manually inactivate the replication consistency group using the terminate command. When the replication consistency group is terminated, volume I/Os bypass the SDR and go directly to the SDS. You can terminate the replication consistency group at both sites separately. Alternatively, you can terminate the replication consistency group at one site; once connectivity is restored, the peer will automatically terminate the replication consistency group at the other site.

Failover can be performed when the replication consistency group is inactive, however consider that the data at the target may be old. While in failover mode, mappings at both the source and target can be activated and volumes can be written to, just like when the replication consistency group is in an active state. There is no data synchronization when the replication consistency group is inactive. You can activate the replication consistency group only once it is no longer in failover mode.

 **NOTE:** Activating a terminated replication consistency group requires full initialization. The initial copy process may take a long time.

The active or inactive replication consistency group state and the commands to terminate or activate a replication consistency group are only available if both peer systems are running PowerFlex v3.6.x or later.

Pausing replication

Replication can be paused for a given replication consistency group.

While paused, any application I/O is stored in the source journal. Once replication is resumed, the journal contents are sent to the target to be applied to the target volumes.

Using replicated volumes

While replication is active, the target volumes remain inaccessible to the hosts with only limited read-only access permitted. This is done to maintain data integrity and consistency between the two peer systems. The data at the target can be accessed to test replication, create snapshots, and to fail over.

Accessing target volumes

There are several situations in which the target volumes are made accessible to the hosts.

- Fail over the replication consistency group, used especially for disaster recovery. During failover, the application I/Os are stopped at the source and the access mode of the source volumes is changed to inaccessible. The target volumes are brought to the newest consistent image available on the destination and their access mode is changed to read-write. Recovery from a failed over state to a normal operation state can be achieved by either restoring the replication, or reversing it.
- Test a failover of the replication consistency group. This allows you to test failover mode, and provides the application with access to the target volumes, without interrupting data synchronization.
- Switch over the replication consistency group. Switchover is a planned failover. The application I/Os are stopped at the source, and the data is synchronized so that the target volume is consistent with the source volume. The access mode of the original source volumes is changed to unavailable or read-only. The access mode of the original target volumes is changed so that the hosts can read and write to the volumes.

Target volumes can be accessed in R/W only after failover or switchover has been performed. Once the replication consistency group is in failover or switchover mode, you can decide how to continue with replication:

- Restore replication—This maintains the replication direction from the original source and overwrites all data at the target.
- Reverse replication—This changes the direction so that the original target becomes the source. All data at the original source is overwritten by the data at the target.

 **NOTE:** Prior to performing the restore or reverse operation, you should consider creating a snapshot on the new target volumes in order to preserve the consistent image during the resynchronization process.

Replication direction and mapping

Use the table as a reference for replication direction and default access to volumes according to a subsequent replication consistency group operation and action of the PowerFlex system.

The replication involves two peers, system A and system B. When the replication is set up, system A is set as source and system B is set as target. The following replication direction refers to the initial direction A to B and to changes to that direction.

Table 22. Replication direction and mapping

Subsequent RCG operations	Possible actions	Replication direction/access	Access to volumes
Normal	Switchover/test failover/ failover Remove	A to B	Access to the volumes is allowed only through the source (system A)
After failover	Reverse/restore Remove	N/A - data is not replicated	By default access to the volume is allowed through the original target (system B) It is possible to enable access through the original source (system A)
After failover + reverse  NOTE: Switchover and test failover are only possible after the peers are synchronized.	Switchover/test failover/ failover Remove	B to A	Access to the volumes is allowed only through the original target (system B)
After failover + restore  NOTE: Switchover and test failover are only possible after the peers are synchronized.	Switchover/test failover/ failover Remove	A to B	Access to the volumes is allowed only through the source (system A)
After switchover	Switchover/test failover stop/failover Remove	B to A	Access to the volumes is allowed only through the original target (system B)
After test failover	Switchover/test failover/ failover Remove	A to B	Access to the volumes is allowed through both systems (system A and system B)

Setting up replication

You are required to initially prepare the infrastructure to use replication in PowerFlex.

The following is an overview of the necessary steps.

1. Add peer systems, at both the source and target systems. Add the corresponding certificates to each system.
2. Add journal capacity, ensuring sufficient capacity.
3. Add SDRs, at least two per protection domain that has replicated volumes.

Perform the following steps to configure replication:

1. Create the replication consistency group. Define the recovery point objective.
2. Add volume pairs. Volumes in a volume pair must have the same size. It is not necessary but is recommended that they have the same attributes (medium or fine granularity, zero padded, or non-zero padded).

If both systems are 4.x or higher then the target volumes may be automatically provisioned when the pair is created.

3. If the replication consistency group was created as active (the default), then initial copy is started automatically. Otherwise, the replication consistency group must be activated before the initial copy is started.

For additional information about replication, see the *Dell PowerFlex and PowerFlex Manager Administration Guide*.

Dell APEX Block Storage

Dell APEX Block Storage for public cloud is a deployment of Dell PowerFlex, software-defined block storage innovation, in the public cloud. APEX Block Storage is available for both AWS and Microsoft Azure allowing you to experience the same benefits of enterprise-class storage services in the cloud as with on-premises. It provides higher performance, larger volume sizes, and improved resilience than what is currently available on the public cloud.

Table 23. Capabilities and benefits

Capability	Benefit or Value
Unique multi-availability zone durability	- Enhanced, space efficient data protection through the aggregation of all instances across availability zones for volume provisioning. - Enables high resilience across availability zones by protecting the data during availability zone failure, without replication of data (space and IOPs).
Extreme performance for mission-critical workloads running in the cloud	Meets and exceeds SLAs with extreme performance (high throughput and low latency) for workloads such as databases and analytics.
Scalable, flexible, and resilient	Meets stringent SLAs and run workloads with confidence and assurance with near linearly scalable performance as workload demand increases.
Data mobility	Seamless data mobility with the ability to easily move data from on-premises to the cloud or across regions within the cloud as workloads demand increases.
Self-healing architecture with rapid rebuild	Ensure high availability and service availability under failure conditions with fast re-protection and rebuild.
Robust ecosystem of automation tools and frameworks (CSM/CSI, Ansible, REST APIs, DDVE, CloudLink, CloudIQ)	Optimal usage and performance of cloud-based block storage with easy to deploy, expand, monitor, and manage capabilities.

NOTE: The following features that are supported by PowerFlex are NOT supported with the APEX Block Storage in the public cloud: compression, fine granularity storage pools in the public cloud, PowerFlex file services, and NVMe over TCP.

APEX Block Storage for AWS

Dell APEX Block Storage for AWS empowers enterprises to run diverse workloads in the public cloud while ensuring extreme performance, scalability, and a simplified cloud experience. APEX Block Storage can be deployed in two configurations by using Elastic Block Store (EBS) volumes, or native NVMe SSD drives attached to EC2 instances (EC2 instance store). Dell APEX Block Storage also provides the proven enterprise data services, such as thin provisioning, snapshots, and asynchronous replication, required to run demanding block-based workloads in the public cloud. Dell APEX Block Storage native asynchronous replication enables data mobility between on-premises and the cloud or across regions in the cloud, for example, AWS East to AWS West.

APEX Block Storage for Microsoft Azure

Now available Dell APEX Block Storage for Public Cloud in Microsoft Azure. APEX Block Storage can be deployed in two configurations using Azure Managed Disks or virtual machines with attached NVMe SSDs based on the use case. Dell APEX Block Storage also provides the proven enterprise data services, such as thin provisioning, snapshots, and asynchronous replication, required to run demanding block-based workloads in the public cloud. Dell APEX Block Storage native asynchronous replication enables data mobility between on-premises and the cloud or across regions in the cloud.

Protection and load balancing

PowerFlex maintains user data in a distributed mesh mirrored layout. Each piece of data is stored on two different fault units. The copies are distributed over the storage devices according to an algorithm that ensures uniform load of each fault unit by terms of capacity and expected network load. Rebuild and rebalance processes are fully automated, but are configurable.

Rebuild

PowerFlex initiates a rebuild process in response to failure. Forward rebuild refers to creating a new copy of the data on another server. Backward rebuild refers to re-synchronizing one of the copies.

When a failure occurs, such as on a server, device or network failure, PowerFlex immediately initiates a process of protecting the data. This process is called *rebuild*, and comes in two flavors:

- *Forward rebuild* is the process of creating another copy of the data on a new server. In this process, all the devices in the storage pool work together, in a many-to-many fashion, to create new copies of all the failed storage blocks. This method ensures an extremely fast rebuild.
- *Backward rebuild* is the process of re-synchronization of one of the copies. This is done by passing to the copy only changes made to the data while this copy was inaccessible. This process minimizes the amount of data transferred over the network during recovery.

PowerFlex automatically selects the type of rebuild to perform. This implies that in some cases, more data will be transferred to minimize the time that the user data is not fully protected.

Rebuild throttling

The rebuild throttling policy determines the priority of rebuild I/Os versus application I/Os when accessing SDS devices. The possible rebuild throttling policies are no limit on rebuild I/Os, limit concurrent I/Os per SDS device, favor application I/Os and dynamic bandwidth throttling.

Rebuild throttling sets the rebuild priority policy for a storage pool. The policy determines the priority between the rebuild I/O and the application I/O when accessing SDS devices. Note that application I/Os are continuously served.

Applying rebuild throttling will increase the time the system is exposed with a single copy of some of the data, but it will reduce the impact on the application. You must attempt to choose the right balance between the two.

The following priority policies may be applied:

- No limit: No limit on rebuilding I/Os. Any rebuild I/O is submitted to the device immediately, without further queuing.

 **NOTE:** Rebuild I/Os are relatively large and hence setting this policy will speed up the rebuild, but will have the maximal effect on the application I/O.

- Limit concurrent I/O: Limit the number of concurrent rebuild I/Os per SDS device (default). The rebuild I/Os are limited to a predefined number of concurrent I/Os. When the limit is reached, the next incoming rebuild I/O waits until the completion of a currently performed rebuild I/O. This will complete the rebuild quickly for best reliability, however, there is a risk of host application impact.
- Favor application I/O: Limit rebuild in both bandwidth and concurrent I/Os. The rebuild I/Os are limited both in bandwidth and in the amount of concurrent I/Os. As long as the number of concurrent rebuild I/Os, and the bandwidth they consume, do not exceed the predefined limits, rebuild I/Os will be served. Once either threshold is reached, the rebuild I/Os wait until both I/O and bandwidth are below their thresholds. For example, setting the value to **1** guarantees the device has one concurrent rebuild I/O at any given moment, which ensures the application I/Os only wait for 1 rebuild I/O at worst case. This imposes bandwidth on top of the limit concurrent I/Os option, which is a prerequisite to using this policy.
- Dynamic bandwidth throttling: This policy is similar to favor application I/O, but extends the interval in which application I/Os are considered to be flowing by defining a minimal quiet period. This quiet period is defined as a certain interval in which no application I/Os occurred. Note that the limits on the rebuild bandwidth and concurrent I/Os are still imposed.
- Default values:
 - The default policy for rebuild is: Limit Concurrent I/O
 - Rebuild concurrent I/O Limit: 1 concurrent I/O

 **NOTE:** Rebuild throttling affects system performance and should only be used by advanced users.

Rebalance

When PowerFlex detects that user data is not balanced across devices in the storage pool, it initiates a process to restore the balance in which data copies are moved from the most used devices to the least used.

Rebalance is the process of moving one of the data copies to a different server. It occurs when PowerFlex detects that the user data is not evenly balanced across the fault units in a storage pool. This can occur as a result of several conditions such as: SDS addition or removal, device addition/removal, or following a recovery from a failure. PowerFlex moves copies of the data from the most used devices to the least used ones.

Both rebuild and rebalance compete with the application I/O for the system resources, which include network, CPU and disks. PowerFlex provides a rich set of parameters that can control this resource consumption. While the system is factory-tuned for balancing between speedy rebuild or rebalance and minimization of the effect on the application I/O, the user has fine-grain control over the rebuild and rebalance behavior.

Rebalance throttling

The rebalance throttling policy determines the priority of rebalance I/Os versus application I/Os when accessing SDS devices.

Rebalance throttling sets the rebalance priority policy for a storage pool. The policy determines the priority between the rebalance I/O and the application I/O when accessing SDS devices. Please note that application I/Os are continuously served. Rebalance, unlike rebuild, does not impact the reliability of the system, and therefore reducing its impact is not risky.

 **NOTE:** Rebalance throttling affects the performance of the system, and should be used only by advanced users.

The following possible priority policies may be applied:

- **No Limit:** No limit on rebalance I/Os. Any rebalance I/O is submitted to the device immediately, without further queuing. Please note that rebalance I/Os are relatively large and hence setting this policy will speed up the rebalance, but will have the maximal effect on the application I/O.
- **Limit concurrent I/O:** Limit the number of concurrent rebalance I/Os per SDS device. The rebalance I/Os are limited to a predefined number of concurrent I/Os. Once the limit is reached, the next incoming rebalance I/O waits until the completion of a currently executed rebalance I/O. For example, setting the value to **1** will guarantee that the device will only have one rebalance I/O at any given moment, which will ensure that the application I/Os only wait for 1 rebalance I/O in the worst case.
- **Favor application I/O:** Limit rebalance in both bandwidth and concurrent I/Os. The rebalance I/Os are limited both in bandwidth and in the amount of concurrent I/Os. As long as the number of concurrent rebalance I/Os, and the bandwidth they consume, do not exceed the predefined limits, rebalance I/Os will be served. Once either limiter is reached, the rebalance I/Os wait until such time that the limits are not met again. This imposes a bandwidth limit on top of the limit concurrent I/Os option.
- **Dynamic bandwidth throttling:** This policy is similar to favor application I/O, but extends the interval in which application I/Os are considered to be flowing by defining a minimal quiet period. This quiet period is defined as a certain interval in which no application I/Os occurred. Note that the limits on the rebalance bandwidth and concurrent I/Os are still imposed.
- **Default values:**
 - The default policy for rebalance: Favor Application I/O
 - Rebalance concurrent I/O Limit: 1 concurrent I/O per SDS device
 - Rebalance bandwidth limit: 10240 KB/s

For information on instant maintenance mode that uses rebuild and rebalance, see [Instant maintenance mode](#).

Checksum protection

PowerFlex calculates and validates the checksum value for the payload during transit to protect data in-flight. Checksum protection is applied to all I/Os.

This feature addresses errors that change the payload during the transit through PowerFlex. PowerFlex protects data in-flight by calculating and validating the checksum value for the payload at both ends.

 **NOTE:**

- The feature is off by default for medium granularity.
- The checksum feature can have a significant impact on large block I/O latency. Contact Dell Technologies Support for more information.

- During write operations, the checksum is calculated when the SDC receives the write request from the application. This checksum is validated just before each SDS writes the data on the storage device.
- During read operations, the checksum is calculated when the data is read from the SDS device, and is validated by the SDC before the data returns to the application. If the validating end detects a discrepancy, it will initiate a retry. The checksum will be done in the granularity of a sector (1/2 KB).
- Pools with fine granularity with or without compression, have a persistent checksum by default. This cannot be changed.

Each I/O goes through compression, and the checksum is calculated before it is written to the disk. There are two types of checksum:

- Fine granularity layout saves checksum data before and after processing to guarantee data integrity (compressed or not).
- There are also system checksums for metadata.

This feature applies to all I/Os: application, rebuild, rebalance, and migrate. The checksum is also kept in RMcachel(read memory cache), protecting every block that is maintained in SDS memory against memory corruption. The checksum feature can be enabled at the protection domain level, and defined at the storage pool level. The feature is T10/DIF-ready.

The fine granularity data layout has a default checksum whether it is compressed or not.

Fine granularity layout checksum implementation

The fine granularity data layout has a default checksum whether it is compressed or not.

Cache

PowerFlex supports RAM Read Cache (using DRAM server memory), to enhance system performance.

The caching modes provided by the system are the following:

- Mode: RAM Read Cache (RMcache).
- Description: Read-only caching performed by server RAM.
- Considerations: RAM Read cache, the fastest type of caching, uses RAM that is allocated for caching. Its size is limited to the amount of allocated RAM.

 **NOTE:** The amount that may be allocated is limited, and can never be the maximum available RAM.

- Default setting: Disabled, except when storage-only nodes are deployed.

The following table illustrates the caching support matrix:

Table 24. Caching support matrix

System	RMcache
PowerFlex	Yes
VxFlex Ready Node with Dell PowerEdge 640, Dell PowerEdge R740xd, Dell PowerEdge 840	Yes

Networking requirements and limitations

In PowerFlex, inter-node communication manages data locations, rebuild and rebalance, and access applications to stored data on one IP network, or on separate IP networks.

Use any of the management interfaces to perform management tasks in the following ways:

- Through a separate network with access to the other PowerFlex components
- On the same network

Configure these options in the following ways:

- During deployment in the full PowerFlex Installer (using the CSV topology file).
- After deployment with the CLI.

The following table includes the list of PowerFlex networking requirements and supported architectures or configurations.

Table 25. Networking requirements and limitations

PowerFlex minimum network requirements and supported architectures	PowerFlex network limitations
• Ethernet-based • Minimum supported speed: 2*10 GbE • Recommended speed: 2*25 GbE • Certified configurations: 4*25 GbE, 2*100 GbE, 4*100 GbE • Supported MTU: 1500/9000 (jumbo frames recommended for optimal performance) • High availability and load-balancing use at least two physical ports with: ◦ PowerFlex native multipath across two or more subsets: ▪ Supports one to eight subnets ◦ Link aggregation: ▪ A single IP subnet can be used if link aggregation is used, otherwise a minimum of two subnets are required. ◦ It is recommended to use MDM virtual IP for simplified network stability if there are failures. • IP addresses: See guidelines about IP address calculation. • Data network latency requirements: ◦ MDM: Maximum of 300 milliseconds • Switch redundancy: ◦ Switches must be redundant. This provides continued access to components inside the rack in the network in case an access switch fails. ◦ Flat network is supported. ◦ Leaf-spine architecture is recommended for cross-rack traffic. ◦ VXLAN is supported. • SDC connection through a router is supported and required for a low latency connection for optimal performance. • PowerFlex has a single management subnet which can be a dedicated port or a VLAN on a trunk connection. For best practices, see: ◦ VxFlex Ready Node R640-R740xd-R840 Operating System Installation and Configuration Guide for Linux ◦ VxFlex Ready Node R640-R740xd-R840 Operating System Installation and Configuration Guide for VMware ESXi	• RMDA is not supported. • InfiniBand is not supported. • IPv4 is supported on PowerFlex software offering. • IPv4 is supported on PowerFlex appliance and PowerFlex rack offerings. • DHCP must not be deployed on a network where PowerFlex MDMs, SDS, SDR, or SDCs reside. • Firewall rules must allow PowerFlex ports. For port information, see the <i>Dell PowerFlex and PowerFlex Manager Security Configuration Guide</i>.

The following table includes the list of PowerFlex networking requirements and supported architectures for systems with replication.

Table 26. Networking requirements and limitations for systems with replication

PowerFlex network requirements and supported architectures for systems with replication	PowerFlex network limits for systems with replication
• Minimum best practice with replication: 4*25 GbE or 2*100 GbE • Additional IP addresses for site to site routing include: ◦ Within a protection domain, SDRs are installed on the same hosts as SDSs. The traffic that an SDR writes to the journal volume is sent to all SDSs that host the journal, not only the one that is co-located on the host. In the backend storage network, each SDR listens on	• MDM to MDM peer metadata synchronization should take place over a WAN with less than 200-millisecond latency. • Local SDR to remote SDR: Latency is not as sensitive in SDR greater than SDR traffic, but round-trip time should not be greater than 200 milliseconds. • Firewall rules must allow PowerFlex ports. For port information, see the <i>Dell PowerFlex and PowerFlex Manager Security Configuration Guide</i>.

Table 26. Networking requirements and limitations for systems with replication

PowerFlex network requirements and supported architectures for systems with replication	PowerFlex network limits for systems with replication
the same node IP addresses as the SDSs and should be able to reach all SDSs within the protection domain. ○ The SDRs require additional, distinct IP addresses which allow them to communicate with remote SDRs. Usually, these should be routable addresses with a properly configured gateway. For redundancy, each SDR should have two. ● Static routes: ○ PowerFlex asynchronous replication usually happens over a WAN between physically remote clusters that do not share the same address segments. If the default route itself is not suitable to properly direct packets to the remote SDR IP addresses, static routes must be configured to indicate either the next hop address or the egress interface or both for reaching the remote subnet. ○ PowerFlex Manager can add static routes for replication use cases.	

See the following configuration guidelines used by VxFlex Ready Node for example purposes only:

- Linux examples: R640 and R740xd nodes see [Configuring network ports on Linux servers](#)
- VMware ESXi examples: R640, R740xd, and R840 nodes see [Configuring network ports on VMware ESXi hosts](#)

This section describes how to choose from these options, depending on the requirements of your organization, security considerations, performance needs, and IT environment.

PowerFlex networking considerations:

- Single IP network: All communications and IOs used for management and for data storage are performed on the same IP network. This setup offers the following benefits:
 - Ease of use
 - Fewer IP addresses required
- Multiple separate IP networks: Separate networks are used for management and for data storage, or separate networks are used within the data storage part of the system. This setup offers the following benefits:
 - Security
 - Redundancy
 - Performance
 - Separate IP roles in order to separate between customer data and internal management

NOTE: Network high availability can be implemented by using NIC-bonding or by using several data networks in PowerFlex.

For more information about MTU performance considerations and best practices, see the *Dell PowerFlex and PowerFlex Manager Install and Upgrade Guide*.

NOTE: The MDM cluster IP address limit is 16 IP addresses, which include all cluster members (primary, secondary, and standby).

Table 27. Range of potential IP address configurations

Column in CSV file	MDM Management IP	MDM IP addresses	SDS all IP addresses	SDS-SDS only IP addresses	SDS-SDC only IP addresses
Comments	Management access	Control network	Rebuild and data path network	Rebuild network	Data path network
	○ Optional, but recommended	○ Mandatory IP addresses used for MDM control	○ IP addresses used for both SDS-SDS and	○ IP addresses used for SDS-SDS	○ IP addresses used for SDS-

Table 27. Range of potential IP address configurations (continued)

Column in CSV file	MDM Management IP	MDM IP addresses	SDS all IP addresses	SDS-SDS only IP addresses	SDS-SDC only IP addresses
	- This is not applicable for tiebreaker IP addresses that can be used to provide access to PowerFlex management applications, such as PowerFlex Manager, CLI, REST API, OpenStack. - This IP address must be externally accessible.	- communications with SDSs and SDCs, used to convey data migration decisions, but no user data passes through the MDM. - Must be on the same network as the data network. - Must be externally accessible if no MDM Management IP addresses are used.	- SDS-SDC communications. - These IP addresses will also be used to communicate with the MDM.	- communication only. - These addresses are used for rebuild and rebalance operations. - These IP addresses will also be used to communicate with the MDM.	- SDC communication. - These addresses are only used for read/write user data operations.

The following combinations can be used for SDS or SDC:

- Only *SDS All IPs*
- Only *SDS-SDS Only IPs* + *SDS-SDC Only IPs*
- SDS All IPs* + either *SDS-SDS Only IPs* or *SDS-SDC Only IPs* (can be used in cases of multiple networks; ensure that you do not use the same IP address more than once in the networks).
- SDS All IPs* + both *SDS-SDS Only IPs* and *SDS-SDC Only IPs* (can be used in cases of multiple networks; ensure that you do not use the same IP address more than once in the networks).

Network limitations

When deploying PowerFlex on VMware ESXi, the same IP subnet can be configured on different VM kernel interfaces and virtual switches.

For more information, see [VMware ESXi limitation](#).

PowerFlex only supports the following network configurations when deploying on VMware ESXi:

- A single data storage network
- Two or more data networks, each on separate IP subnets
- A single IP address data network using several NIC-bonding configurations, or vSwitch load-balancing

Network stability

Resilient network connectivity is standard for PowerFlex. Nevertheless the actual implementation may vary. In some cases, resiliency is attained through LACP or bonding, while in other cases through application-level multipathing.

PowerFlex has additional capabilities to support partial failures on existing connections. While many network issues are manifested as a complete disconnection, some are not. PowerFlex is aware of degraded connections, including:

- Stable, but slow connections in relation to others connections.
- Unstable connections (occasional disconnections).
- A combination of the above two.

Additional PowerFlex capabilities support the following error scenarios:

- PowerFlex contains safeguards to maintain the proxy connection between the SDC and SDS. In the PowerFlex architecture, each SDC is required to be fully connected to all the SDSs in each of the protection domains from which it consumes volumes. In cases where the SDC is unable to reach one or more of the SDSs, the MDM reflects that and the management

API raises an alert. The SDC then sends the I/O to another SDS within the same protection domain, or subset of this protection domain. The new SDS is referred to as a proxy SDS, because its only role is to forward the I/O to the SDS which owns the data, receive the I/O result, and return the result to the initiating SDC.

- PowerFlex supports using the MDM secondary IP addresses as a proxy connection to the primary MDM. This capability is applicable only when MDM VIP is not in use, because it requires configuration of all the SDC IP addresses to attempt a proxy connection.

Networking requirements for PowerFlex file deployment

The following lists the networking requirements for a PowerFlex file node.

- Ensure that a NIC with an IP address is configured in the same range as the following networks:
 - PowerFlex management
 - PowerFlex data (1 to 4)
- Create a bonded interface on the PowerFlex node. No sub interfaces or IP addresses should be assigned to this bonded interface dedicated to NAS traffic.
- PowerFlex software and PowerFlex custom node support a single NIC or logical interface for PowerFlex file services without the bonding requirement. For more details, see the *Dell PowerFlex and PowerFlex Manager Install and Upgrade Guide*.
 - This use case enables customers with existing network infrastructure to leverage the PowerFlex file offering. Dell Technologies recommends having a high-availability network of some form.

Sample configurations

A sample network configuration:

```
eth0 - PowerFlex Management (Node management)
eth1 - PowerFlex Data
eth2 - PowerFlex Data
bond0 (there should not be any tagged interface under bond0)
```

A sample IP address configuration on the PowerFlex file node:

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen
1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen
1000
    link/ether 00:50:56:b0:95:da brd ff:ff:ff:ff:ff:ff
    altname enp11s0
    altname ens192
    inet 10.1.1.226/20 brd 10.234.223.255 scope global eth0
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen
1000
    link/ether 00:50:56:b0:bb:8c brd ff:ff:ff:ff:ff:ff
    altname enp19s0
    altname ens224
    inet 172.16.221.226/16 brd 172.16.255.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::250:56ff:feb0:bb8c/64 scope link
        valid_lft forever preferred_lft forever
4: eth2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen
1000
    link/ether 00:50:56:b0:3f:09 brd ff:ff:ff:ff:ff:ff
    altname enp27s0
    altname ens256
    inet 172.17.221.226/16 brd 172.17.255.255 scope global eth2
        valid_lft forever preferred_lft forever
    inet6 fe80::250:56ff:feb0:3f09/64 scope link
        valid_lft forever preferred_lft forever
5: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group
default
```

```

    link/ether 02:42:14:01:33:16 brd ff:ff:ff:ff:ff:ff
    inet 172.29.200.1/24 brd 172.29.200.255 scope global docker0
        valid_lft forever preferred_lft forever
6: eth3: <BROADCAST,MULTICAST,SECONDARY,UP,LOWER_UP> mtu 1500 qdisc mq primary bond0
state UP group default qlen 1000
    link/ether 00:50:56:b0:65:04 brd ff:ff:ff:ff:ff:ff
    altname enp12s0
    altname ens193
7: eth4: <BROADCAST,MULTICAST,SECONDARY,UP,LOWER_UP> mtu 1500 qdisc mq primary bond0
state UP group default qlen 1000
    link/ether 00:50:56:b0:65:04 brd ff:ff:ff:ff:ff:ff
    altname enp4s0
    altname ens161
8: bond0: <BROADCAST,MULTICAST,PRIMARY,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP
group default qlen 1000
    link/ether 00:50:56:b0:65:04 brd ff:ff:ff:ff:ff:ffa

```

NOTE: The switch port that is connected to interfaces that are assigned to NAS bond interface should be configured to accept NAS management VLAN as untagged and NAS data VLANs as tagged.

A sample Cisco Nexus switch configuration:

```

interface ethernet1/1/4
    no shutdown
    switchport mode trunk
    switchport access vlan 100
    switchport trunk allowed vlan 200,201
    mtu 9216
    flowcontrol receive off
    channel-group 8 mode active
interface port-channel10
    switchport mode trunk
    switchport trunk native vlan 100
    switchport trunk allowed vlan 200,201
    mtu 9216
    no lacp suspend-individual
    lacp vpc-convergence vpc 8

```

Hosts

PowerFlex supports both storage data clients (SDCs) and NVMe hosts to map and present volumes to the operating system. SDCs and NVMe hosts can co-exist in the same cluster; the only restriction is that they cannot use the same volume simultaneously.

The SDC is a PowerFlex-proprietary lightweight device driver that provides frontend volume access to applications, and maps and presents volumes to the operating system or hypervisor. SDCs maintain peer-to-peer connections to every SDS managing a pool of storage. They translate SCSI to PowerFlex protocol.

NVMe hosts consume PowerFlex block storage using NVMe over Fabrics (NVMe-oF) connectivity to map and present volumes to the operating system or hypervisor. From a user perspective, NVMe hosts play the same role in the system as SDCs. However, they co-exist on the SDS and do not require any additional hardware.

The following figures show the difference in connectivity between the SDC and the NVMe host. The SDC sits on the application host or compute node and connects directly to the storage layer. The NVMe host, on the other hand, connects to an NVMe target called the storage data target, which runs on the storage side. The storage data target directs the volumes onward to the storage components.

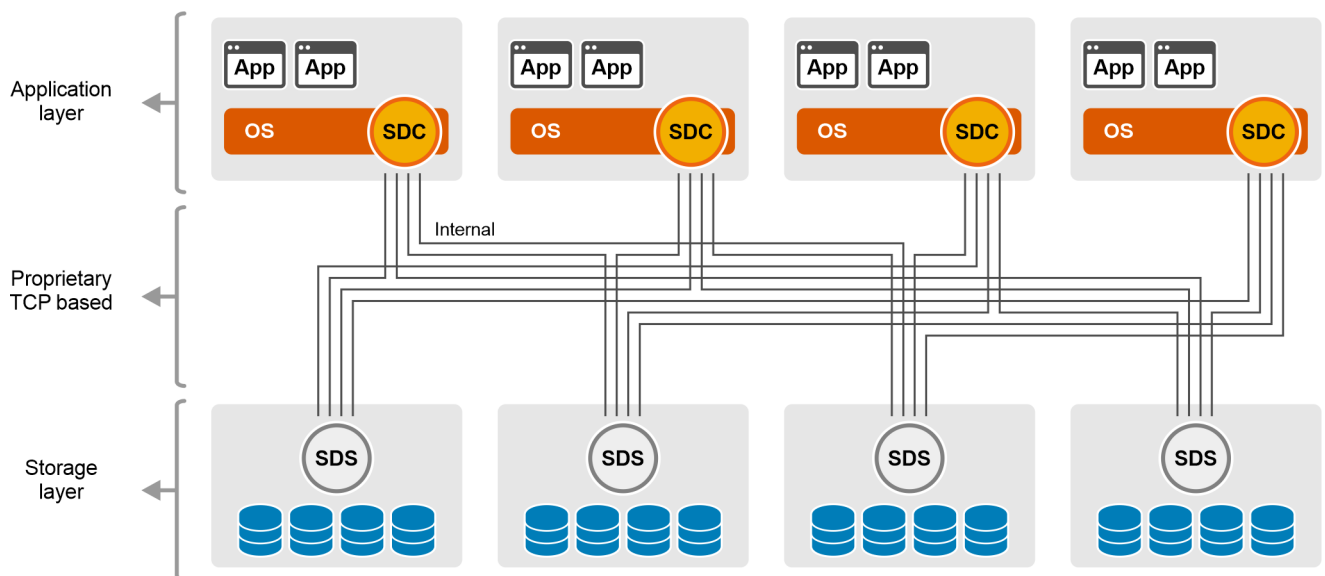


Figure 4. SDC host connectivity

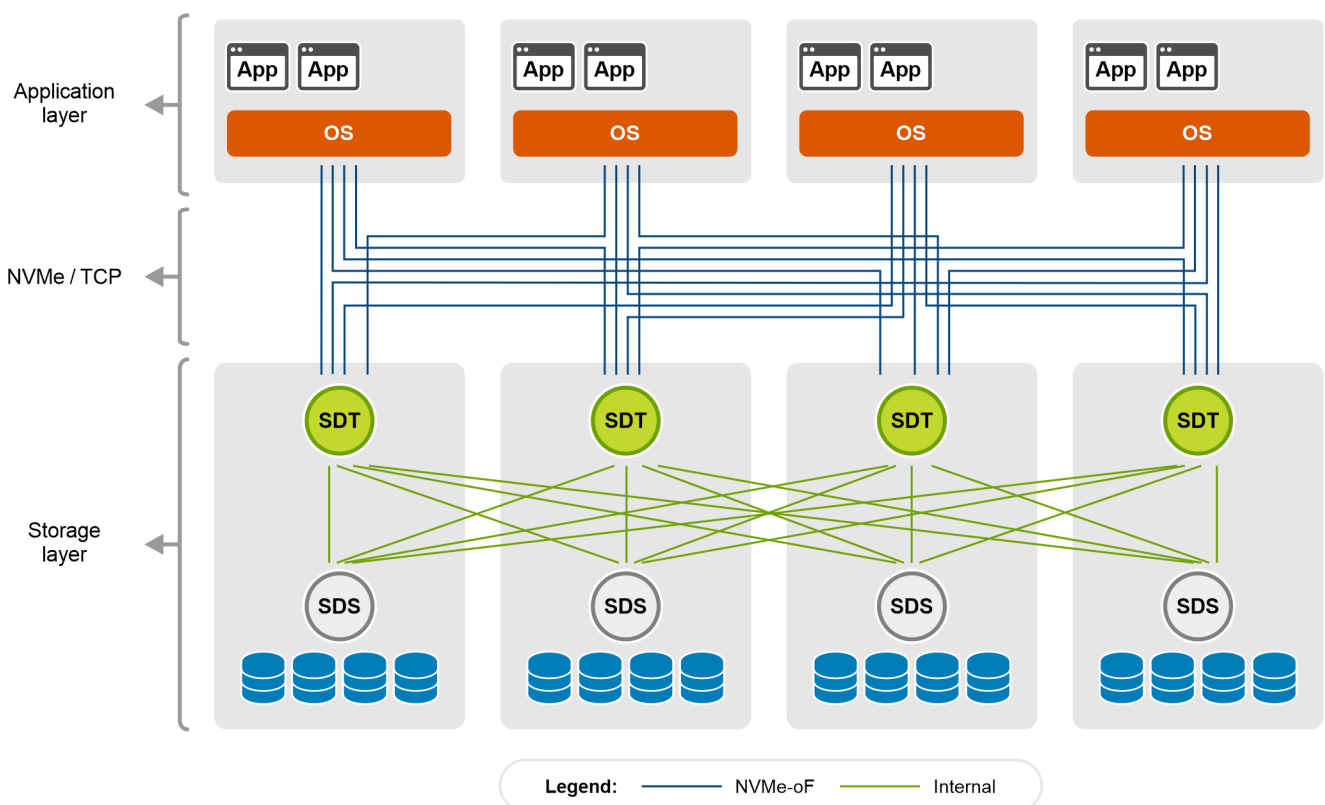


Figure 5. NVMe over TCP host connectivity

Storage data clients

The storage data client is a driver installed on hosts that translates the PowerFlex control and data path protocols to the SCSI layer. The SDC handles all the PowerFlex multipathing aspects and connects to all the PowerFlex storage components.

The SDC also integrates with relevant operating system logic to provide a native block storage device in the operating system. Volumes (also known as LUNs in other storage systems) are mapped to the SDC to be consumed as block devices.

SDCs are supported on Linux, Windows, AIX, and other operating systems, and can be installed on any non-PowerFlex host meeting PowerFlex specifications for SDCs. PowerFlex appliance and PowerFlex rack also support installing SDCs in hyperconverged configurations.

SCSI support

The SCSI identity consists of the following:

- The ESXi SDC registers a new SCSI adapter with Fibre Channel transport.
- The Fibre Channel node worldwide name is the low 64 bits of the SDC GUID.
- The Fibre Channel port worldwide name is the high 64 bits of the SDC GUID.
- The Fibre Channel port and node worldwide names are also exposed for the virtual Fibre Channel targets.
- Every mapped volume contains the following:
 - A SCSI worldwide name which can be queried at a vital product data (VPD) page 0x83 EUI-64 designator using the SCSI Inquiry command. The EUI-64 designator is 128 bits, and it is composed of the PowerFlex system MDM ID and Volume ID.
 - A SCSI unit serial number can be queried through a VPD page 0x83 EUI-64 designator using the SCSI Inquiry command. The serial number is made of EMC - MDM_ID-Vol_ID
- ```
MOS_SNPRINTF((char *)&pUnitSerial->serialNum[0],
serialNumBufSize, "%s-%.16"MOS_X64_BIT__STRING"-%.16"MOS_X64_BIT__STRING,
MOS_VERSION__LEGACY_COMPANY_NAME_CAP, mapVol_GetMdmID(pVol), mapVol_GetVolID(pVol));
```
- Every mapped volume is assigned a Target LUN address when it is mapped. Because the storage I/O (SIO) volume ID is 64 bits, the address is calculated as follows:
  - The SDC exposes up to 8 K LUNs in total. Begin with the volume index (low 32 bits of the volume ID) and use the modulo operation with 8 K. The result is called the SCSI volume index.
  - The low eight bits of the SCSI volume index become the LUN ID, and the remaining top five bits becomes the target ID. If there is a collision, which is when a volume exists with a TargetLun pair, the LUN ID stays a low eight bits and the target ID increments by one until the collision is resolved. If a collision exists for all 32 target IDs, the LUN ID increments by one and starts over until the collision is resolved.
  - SDC reports a single target worldwide name that is based on the SDC GUID. This target ID is unrelated to the target that is part of the volume LUN ID.

## Supported SCSI commands

The following tables describe the SCSI commands that PowerFlex supports.

**Table 28. Mandatory SCSI commands**

| Command type          | Operation code                                                             | Comments                                                                                           |
|-----------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| FORMAT UNIT           | 04h                                                                        | --                                                                                                 |
| INQUIRY               | 12h                                                                        | INQUIRY commands support a subset of the standard pages.                                           |
| READ                  | READ (10): 28h<br>READ (12): A8h<br>READ (16): 88h<br>READ (32): 7Fh/0009h | PowerFlex supports various sizes of READ commands: READ (10), READ (12), READ (16), and READ (32). |
| READ CAPACITY (10)    | 25h                                                                        | --                                                                                                 |
| READ CAPACITY (16)    | 9Eh/10h                                                                    | --                                                                                                 |
| REPORT LUNS           | A0h                                                                        | --                                                                                                 |
| TEST UNIT READY (TUR) | 00h                                                                        | --                                                                                                 |

**Table 29. Optional SCSI commands**

| Command type                             | Operation code | Comments                                                    |
|------------------------------------------|----------------|-------------------------------------------------------------|
| COMPARE AND WRITE                        | 89h            | Atomic test and set (ATS)                                   |
| MODE SENSE - only page 0xa / subpage 0x0 | 1Ah            | MODE SENSE commands support a subset of the standard pages. |
| <b>Reservations related</b>              |                |                                                             |

**Table 29. Optional SCSI commands (continued)**

| Command type           | Operation code                                                           | Comments                                                                                                                                 |
|------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| PERSISTENT RESERVE IN  | 5Eh                                                                      | SCSI-3 reservation support                                                                                                               |
| PERSISTENT RESERVE OUT | 5Fh                                                                      | SCSI3 reservation support'                                                                                                               |
| RESERVE                | 0x16                                                                     | SCSI-2 reservation support. PowerFlex supports RESERVE 1 and RESERVE 6.                                                                  |
| RELEASE                | 0x17                                                                     | SCSI-2 reservation support. PowerFlex supports RELEASE 6 and RELEASE 10.                                                                 |
| UNMAP                  | 42h                                                                      | Thin provisioning or block delete                                                                                                        |
| WRITE                  | WRITE (10): 2Ah<br>WRITE (12): AAh<br>WRITE (16): 8Ah<br>WRITE (32): 7Fh | PowerFlex supports different sizes of WRITE commands: WRITE(10), (WRITE 12), WRITE (16), and WRITE (32)                                  |
| WRITE SAME             | WRITE SAME (10): 41h<br>WRITE SAME (16): 93h<br>WRITE SAME (32): 7Fh     | Zero block or write zeroes. PowerFlex supports different sizes of WRITE SAME commands: WRITE SAME (10), WRITE SAME (16), WRITE SAME (32) |

The following command is not supported:

- EXTENDED COPY

## Supported VPD pages

PowerFlex supports the following pages for SCSI Vital Product Data (VPD):

- 0x00 - Standard inquiry
- 0x80 - Unit serial number
- 0x83 - Device identification
- 0x80 - Block limits
  - This page includes size limitations used by the optional commands: Unmap, Write Same, and Compare and Write.
- 0x83 - Logical block provisioning

## SDC connection monitoring

The system monitors all connections between SDCs and SDSs and sends out an alert when an active connection between an SDC and an SDS goes down.

To monitor SDC and SDS connections effectively, the MDM collects connectivity updates from all of the SDCs. The MDM posts events whenever an SDC connects to or disconnects from a specific SDS IP address. The MDM frequently analyzes the connectivity status to determine the current system state. The system does not send out alerts for temporary connectivity issues that are resolved in less than 10 seconds.

The following are the possible connectivity states between SDCs and SDSs in the system:

- All connected
- One SDC is disconnected from one SDS
- One SDC is disconnected from one SDS IP address
- One SDC is disconnected from all SDSs
- All SDCs are disconnected from one SDS
- All SDCs are disconnected from one SDS IP address
- All SDCs are disconnected from all SDSs
- Multiple disconnections

When the connectivity state of the system changes to any state other than All Connected, an alert is displayed in PowerFlex Manager and is written to the MDM event log. Once an alert is generated, you can use the SCLI to query details on the disconnection using the command `scli --query_sdc_to_sds_disconnections`.

The MDM does not monitor the connectivity state of SDCs or SDSs in the following scenarios:

- SDS is in maintenance mode
- SDS is disconnected from the MDM
- SDS is in the process of being removed
- SDC is disconnected from the MDM for more than two minutes
- SDC is not approved

## NVMe over TCP overview

NVMe over TCP is based on NVMe over Fabrics (NVMe-oF), which is an industry-standard protocol for storage transport. NVMe-oF is supported by multiple operating systems and does not require any host agent on the storage node.

TCP provides a standard, common, interoperable solution that delivers excellent performance. The TCP transport allows the use of existing networks and hardware.

PowerFlex hosts can use standard NVMe over TCP operating system drivers to connect to PowerFlex. They do not require proprietary host agents. These drivers are available in mainstream Linux distributions and in VMware ESXi.

NVMe over TCP is supported in a disaggregated, two-layer PowerFlex deployment architecture. The NVMe host is deployed on a compute node and communicates directly with the NVMe target, known as the storage data target (SDT), on the storage node. NVMe hosts are deployed through PowerFlex Manager.

PowerFlex supports NVMe over TCP on the following operating systems:

- VMware ESXi 7.0U 3f (Dell Technologies recommends using the latest release) and VMware ESXi 8.0 (Dell Technologies recommends using the latest release)
- SUSE Linux Enterprise Server, SLES 15 SP5, with the x86\_64 with 5.14.21-150500.55.36-default.

All other Linux operating systems are in technology preview state and can be run in testing scenarios.

**NOTE:** Technology preview features are not fully supported and may not be functionally complete. These features are not suitable for deployment in production. However, these features are provided to the customer as a courtesy and the primary goal is for the features to gain wider exposure with the goal of full support in the future. Customers may find these features useful in a nonproduction environment. Customers are encouraged to provide feedback and functionality suggestions for a technology preview feature before it becomes fully supported.

For the most updated list, see the PowerFlex [support matrix](#).

PowerFlex supports NVMe reservations (SCSI-3 equivalent) with NVMe over TCP.

## NVMe hosts

PowerFlex supports connecting and consuming storage from NVMe hosts. An NVMe host is a host running an operating system that supports NVMe over TCP.

After an NVMe host is created in PowerFlex, you can map volumes to the host so that the NVMe host can consume storage. The NVMe host connects to an NVMe target called the storage data target, which runs on the storage side. The storage data target directs the volumes onward to the storage components.

The following host parameters are configurable:

- Number of connections. For hosts that are expected to generate high I/O load, you can increase the default value.
- Number of paths per volume. Up to eight paths can be configured per volume in the host attribute. This is mostly an operating system limit per volume and per total paths.

PowerFlex supports specifying networks that might fail together. NVMe over TCP connections are allocated to the NVMe over TCP hosts through multiple network sets, which offers resiliency to the failure of a network set.

NVMe over TCP supports fault sets. Connections allocated to the NVMe over TCP host are through targets on multiple fault sets that provide path resiliency to a fault set failure.

## Storage data target (NVMe target)

The storage data target (SDT) is the channel for communications with NVMe hosts. It serves as the NVMe target and translates NVMe over Fabrics protocol into internal PowerFlex protocol.

The storage data target is deployed with the SDS on each storage server and provides access to the volumes inside that protection domain.

In PowerFlex, every storage data target provides a discovery service. The information returned to the host includes the ports assigned to that host, either by the automatic load balancing or by manual connectivity configuration.

## Persistent discovery

PowerFlex supports persistent discovery for NVMe over TCP targets. When NVMe hosts are initially discovered through the persistent discovery controller, they remain connected to the discovery service. This enables automatic updates of the hosts regarding storage side changes.

When a change in the discovery information is provided to the NVMe host, the discovery controller returns an asynchronous event notification and the host requests the updated discovery log page.

Changes in discovery information include the following:

- A new volume is mapped to the host from a new protection domain.
- A new storage data target is added to the system.
- Load balancing has moved the host connection from one storage data target to another storage data target.

When using this functionality, confirm that the host operating system supports the persistent discovery controller and that it is enabled for the host. This feature is supported with VMware ESXi 8.0 and Red Hat Enterprise Linux 9, SLES 15.4, and others. For more information, see the NVMe over TCP host configuration in the documentation of the respective operating system.

## Load balancing

PowerFlex supports NVMe over TCP connectivity and adds NVMe-oF targets to enable hosts to connect to storage. When hosts are allowed to connect and consume storage, the system ensures that it remains resilient, balanced, and available.

### PowerFlex NVMe over TCP load balancer

Utilize the optimal PowerFlex system performance by distributing the NVMe host workloads over the NVMe targets (storage data targets). PowerFlex offers two ways to achieve load distribution: manual and automatic. In both cases, system ports are selected for the host and are returned to the host when it performs NVMe discovery.

The automatic load balancing, which is the default policy uses an algorithm that selects the system ports based on the following:

- Access to the mapped volumes of the host
- Path resiliency
- Load balancing

The host connectivity plan is recalculated under the following conditions:

- Host connects.
- Storage data target is added or removed.
- Network set or system network is modified.
- Storage data target is added or removed to a fault set.
- Storage data target IP address is reassigned to a different network.

 **NOTE:** The load balancer ignores temporary issues such as network failures or if a PowerFlex storage-only node goes down for a few minutes.

The storage system provides multiple ports through which the host can connect, though every port and node has a limited throughput. To achieve the best performance, ensure that the combined I/O load of the hosts is distributed well over the available system ports and nodes. The NVMe over TCP load balancer aims to automate the planning and execution of the host-to-storage connectivity task to achieve path resiliency and workload balance.

The PowerFlex load balancer considers the host networks for resiliency. When a host loses a network, it can still use other paths through available networks to access the storage.

## NVMe-oF discovery

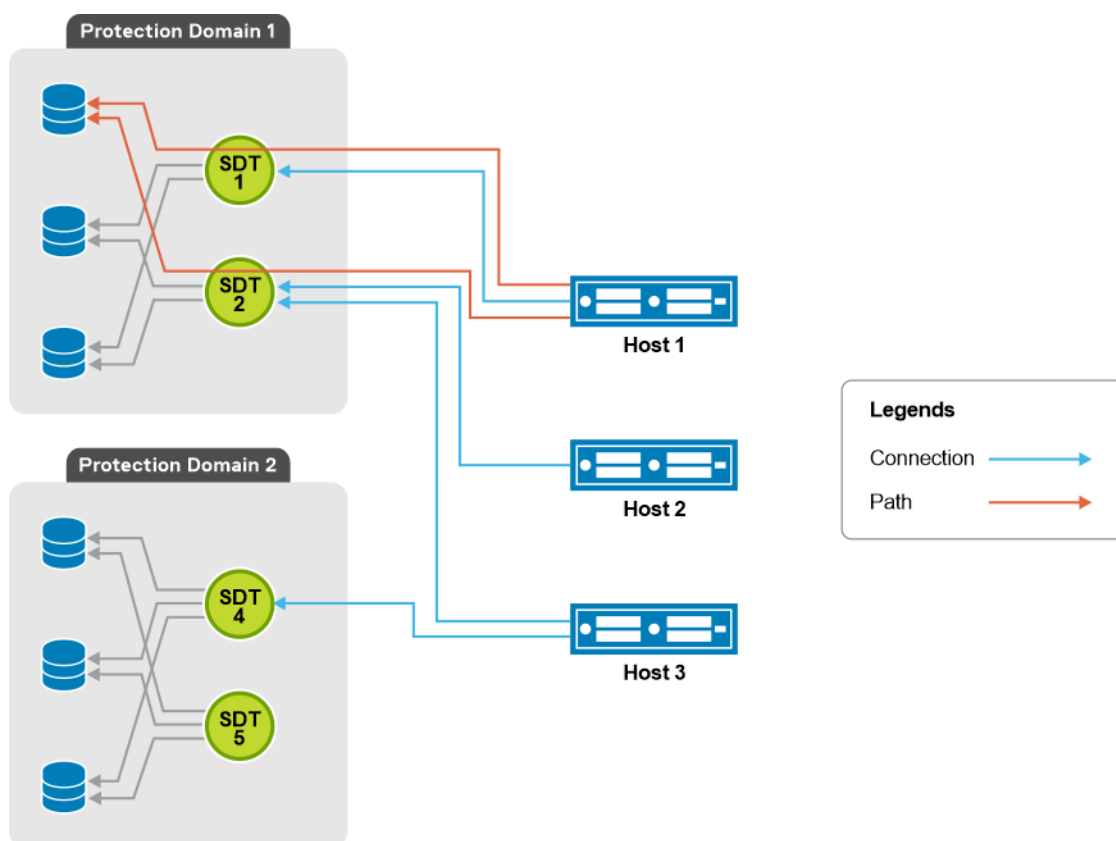
NVMe over Fabrics includes a discovery protocol, which is a standard method for the host to discover and connect to available subsystems.

The host connects to a discovery service and receives discovery information, including all the connection ports to access its volumes. PowerFlex continues to update the host regarding changes using the persistent discovery controller. In PowerFlex, every storage data target provides a discovery service. The information returned to the host includes the ports that are assigned to that host by the load balancer. The host can also be configured manually. If the user opted for the manual connectivity configuration.

## Balancing host connections

When a host is mapped to the first volume from a protection domain, it assigns ports that might connect to volumes in the protection domain. Each port represents a connection. The number of ports that are assigned is a configurable host attribute, which is up to 128 per protection domain. The system selects the least occupied ports to ensure fault isolation. The number of paths per volume is a configurable host attribute, which is a maximum of eight paths per volume. This configurable host attribute is mostly an operating system limit per volume and total paths.

## NVMe over TCP connections and paths



A connection is the establishment of the NVMe controller and includes the following:

- Link between a host part and a specific storage data target IP address.
- Connection couple (host-port and system-port) .
- The load balancer provides 10 ports for each host by default, and up to 128 ports.

A path is a connection that is used to provide access to a volume. This connection makes the path a tuple (host-port, system-port, and volume), which must be a subset of the connections.

By default, the load balancer provides four paths per volume to each host, which is selected from the assigned ports.

## Manual load balancing

Users might prefer to manually configure host connectivity. Manual configuration is appropriate for special cases, such as when hosts generate a higher-than-expected traffic load.

To configure manually, the user designates in the host configuration the system ports to which the host must connect.

## Migration to NVMe over TCP

PowerFlex provides the option to migrate workloads from SDC to NVMe over TCP on VMware ESXi: the migration can be performed either online or offline.

- Online migration using storage vMotion (VMFS only) : The standard way of moving storage is with storage vMotion. Storage vMotion also supports switching protocols by migrating to a new datastore. For more information, see the *Dell PowerFlex Administration Guide* for your PowerFlex version.
- Offline migration (VMFS only): There is a new option for converting an existing VMFS datastore from SCSI (SDC) to NVMe over TCP without sending all the data over the network. For more information about the offline migration steps, see [SCSI to NVMe VMware VMFS Datastore Offline Migration Steps](#).

 **NOTE:** There are no standard ways to migrate Linux environments, VMware ESXi clusters, and RDMs.

## SMART hardware monitoring

The PowerFlex bare-metal solution provides monitoring capabilities for RAID controllers and storage devices compatible with SMART (Self-Monitoring, Analysis and Reporting Technology) protocols.

In Linux-based environments, SMART-compatible HDDs, SSDs and RAID storage controllers can be monitored for SMART attributes such as temperature, SSD wear level, and error counters. LEDs can also be lit on these hardware devices, to simplify physical identification for maintenance purposes.

Each hardware vendor defines specific thresholds for the SMART attributes. This feature currently supports storage devices controlled by LSI, HP and Dell RAID controllers, and stand-alone devices. During system deployment, an external monitoring tool is installed as part of the LIA on each node. Additional RAID controller tools must be installed manually after system deployment: storcli for LSI RAID controllers, hpssacli for HP RAID controllers, or perccli for Dell RAID controllers. These tools are used by the system to collect the counters that are returned to the MDM.

 **NOTE:** In some cases, LSI RAID controllers may report vendor information as "AVAGO" instead of LSI.

The MDM queries the SDSs at set intervals, and stores the returned information. This information can be viewed using CLI queries. In addition, when thresholds are crossed for SMART attributes, alerts are generated by the system.

When the CLI is used to query device information, physical device information, such as serial number, disk slot information, model name, vendor etc., temperature, and wear level information (for SSDs only) is included in the returned response.

You can use PowerFlex Manager to monitor SMART-related alerts in the **Alerts** view.

In addition, SNMP traps and Secure Connect Gateway alert codes can be used to monitor alerts triggered by devices compatible with SMART.

## List of approved RAID controllers

High-level specifications of RAID controllers, which are tested and certified by PowerFlex.

**Table 30. PowerFlex-certified RAID controllers**

| Manufacturer | Specifications                                                                                                                                                                                            |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dell         | <ul style="list-style-type: none"><li>• Model Name: PERC H730 Mini</li><li>• Firmware Version: 25.5.9.0001</li><li>• Driver Version: 07.700.52.00 and above</li><li>• Driver Name: megaraid_sas</li></ul> |
|              | <ul style="list-style-type: none"><li>• Model Name: H755</li><li>• Firmware Version: See VxFlex Ready Node Firmware and Driver support matrix.</li></ul>                                                  |

**Table 30. PowerFlex-certified RAID controllers (continued)**

| Manufacturer | Specifications                                                                                                                                                                                                                                                                   |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <ul style="list-style-type: none"> <li>Driver Version: See VxFlex Ready Node Firmware and Driver support matrix.</li> <li>Driver Name: megaraid_sas</li> </ul>                                                                                                                   |
|              | <ul style="list-style-type: none"> <li>Model Name: H740p</li> <li>Firmware Version: See VxFlex Ready Node Firmware and Driver support matrix.</li> <li>Driver Version: See VxFlex Ready Node Firmware and Driver support matrix.</li> <li>Driver Name: megaraid_sas</li> </ul>   |
|              | <ul style="list-style-type: none"> <li>Model Name: HBA355i/355</li> <li>Firmware Version: See VxFlex Ready Node Firmware and Driver support matrix.</li> <li>Driver Version: See VxFlex Ready Node Firmware and Driver support matrix..</li> <li>Driver Name: mpt3sas</li> </ul> |
|              | <ul style="list-style-type: none"> <li>Model Name: HBA330</li> <li>Firmware Version: See VxFlex Ready Node Firmware and Driver support matrix.</li> <li>Driver Version: See VxFlex Ready Node Firmware and Driver support matrix.</li> <li>Driver Name: mpt3sas</li> </ul>       |
| HP           | <ul style="list-style-type: none"> <li>Model Name: Smart Array P440ar</li> <li>Firmware Version: 3.56</li> <li>Driver Version: 3.4.10</li> <li>Driver Name: hpsa</li> </ul>                                                                                                      |

## NVDIMM hardware awareness feature

From PowerFlex, the customer can enable the NVDIMM hardware awareness feature on the SDS.

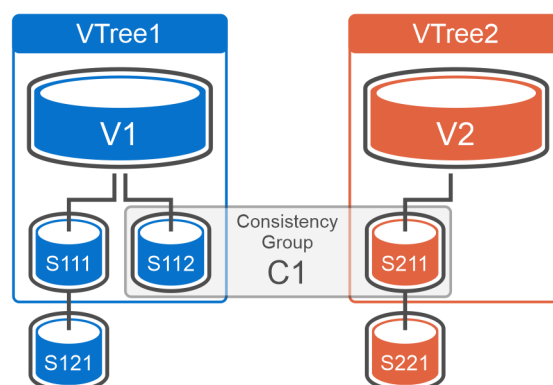
When this feature is enabled, the SDS detects bad blocks on NVDIMM devices used for fine granularity acceleration and takes appropriate actions based on how and when the corruptions in the NVDIMM devices are detected.

## Snapshots

PowerFlex enables you to create snapshots, that is, instantaneous copies of volumes. A snapshot can be manipulated as any volume exposed by the storage system. Snapshots that are taken simultaneously form a consistency group and can be manipulated as a group.

The PowerFlex storage system enables you to take snapshots of existing volumes, up to 128 per volume. The snapshots are thin provisioned and are extremely quick. For more information about thin provisioning, see *SAN virtualization layer* in this guide.

Once a snapshot is generated, it becomes a new, unmapped volume in the system. You can manipulate it in the same manner as any other volume exposed by the PowerFlex storage system.



**Figure 6. Snapshot operations**



The structure related to all the snapshots resulting from one volume is referred to as a vTree (short for volume tree). When taking a snapshot in the system, you can specify more than one volume. All snapshots that are taken together form a consistency group. They are consistent in the sense that they were all taken simultaneously. So if there is a contextual relationship between the data that is contained by all the snapshot members, then that set is meaningful. The consistency group allows manipulation of the entire set.

If you remove an entire consistency group, all snapshots that were taken together are removed. In RED, S211 is a snapshot of V2. Since S112 and S211 were taken together, they compose a consistency group that is designated as C1.

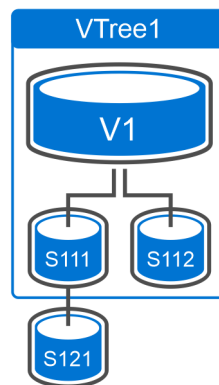
#### **NOTE:**

- The consistency group is only for convenience purposes. There are no protection measures done by PowerFlex to conserve the consistency group. For example, you can remove a snapshot that is a member of a consistency group.
- Support of removal of a parent snapshot in the system. You can merge the parent to a child snapshot.
- It is recommended not to trim volumes that contain snapshots as the logical capacity may not shrink.

## Volume trees (vTrees)

A vTree consists of the root volume and all of its descendant volumes and snapshots.

A vTree is the structure comprised of a volume and the snapshots resulting from that volume. It is a tree spanning from the source volume at its root, whose descendants are either snapshots of the volume itself or snapshots of a snapshot. In the vTree diagram, S<sub>111</sub> and S<sub>112</sub> are snapshots of V<sub>1</sub>. S<sub>121</sub> is a snapshot of snapshot S<sub>111</sub>. Together, V<sub>1</sub> and S<sub>1xy</sub> are the vTree of V<sub>1</sub>.



**Figure 7. vTree diagram**

## Snapshot policies

Snapshot policies enable you to define policies where you can configure the number of snapshots to take at a given time for one or more volumes.

Snapshots are taken according to the rule defined. You can define the time interval in-between two rounds of snapshots as well as the number of snapshots to retain, in a multi-level structure. For example, take snapshots every x minutes/hours/days/weeks. There are one to six levels, with the first level having the most frequent snapshots.

Example:

Rule: Take snapshots every 60 minutes

Retention Levels:

- 24 snapshots
- 7 snapshots
- 4 snapshots

After defining the parameters, select the source volume to add to the snapshot policy. You can add multiple source volumes to a snapshot policy, but only a single policy per source volume is allowed. Only one volume per vTree may be used as a source volume of a policy (any policy). When you remove the source volume from the policy, you must choose how to handle snapshots. Snapshots created by the policy are referred to as auto snapshots. In PowerFlex Manager, this is indicated if a snapshot policy is displayed for the snapshot.

- If the source volume has no auto snapshots, you cannot unassign the source volume from the snapshot policy. You can remove auto snapshots from **Snapshots** in PowerFlex Manager.
- If the source volume has auto snapshots but none of them are locked, you are prompted to confirm that you would like to delete all auto snapshots. If any of the auto snapshots are locked, the locked auto snapshots are just detached from the snapshot policy, but not deleted. You need to manually remove auto snapshots. It does not matter if the auto snapshot is locked or not, you can still delete.

## Performance

PowerFlex is a scale-out, high-performance storage system. The performance of the system is almost perfectly scalable, as shown in the graph. This means that adding nodes adds capacity and also increases the performance of the system.

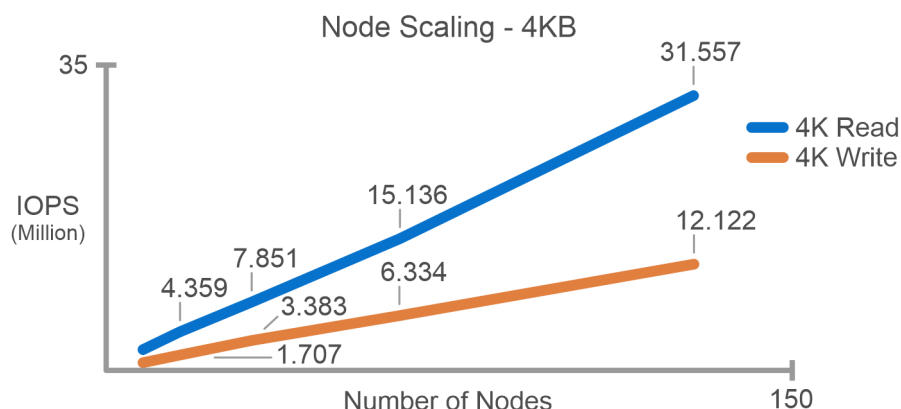


Figure 8. PowerFlex performance

## Migrating volumes

You can migrate volumes to a different storage pool or protection domain within your PowerFlex system.

You can migrate volumes from the following storage layers:

- From one storage pool to another within the same protection domain
- From one storage pool to another across different protection domains
- From the volume tree and all of its snapshots when they are migrated together
- From thick to thin
- From fine or medium to medium or fine granularity

Requirements to migrate volumes:

- Medium granularity pools should have zero padding enabled.

## Implementing PowerFlex

Implementing PowerFlex is, in general, a two-step process: first build the physical storage layer, then configure the virtual SAN layer on top of it.

Communication is done over the existing LAN data network using standard TCP over IP. The MDM and SDS nodes can be assigned to up to eight IP addresses, enabling wider bandwidth and better I/O performance and redundancy.

To build the physical storage layer:

1. Physically assemble the server nodes in the rack and perform all cabling, according to the instructions for your PowerFlex product.
2. Configure the operating system, including all required IP addresses, based on network best practice guidelines.
3. Using PowerFlex Installer, deploy the PowerFlex software.

The physical layer is now ready, and you can expose a virtual storage layer.

PowerFlex appliance and PowerFlex rack offer a more automated installation experience using PowerFlex installer. For more information regarding PowerFlex Manager management and operation capabilities, see PowerFlex appliance and PowerFlex rack product information.

## PowerFlex with Nutanix Cloud Platform

Nutanix Cloud Platform (NCP) for PowerFlex is powered by Acropolis Hypervisor (AHV) for compute nodes and PowerFlex for storage nodes.

Nutanix Cloud Platform is traditionally a hyperconverged platform that consolidates computing, storage, and networking into a unified pool. It uses the Acropolis Operating System (AOS) for storage, AHV for virtualization, and Prism Central for centralized management across environments.

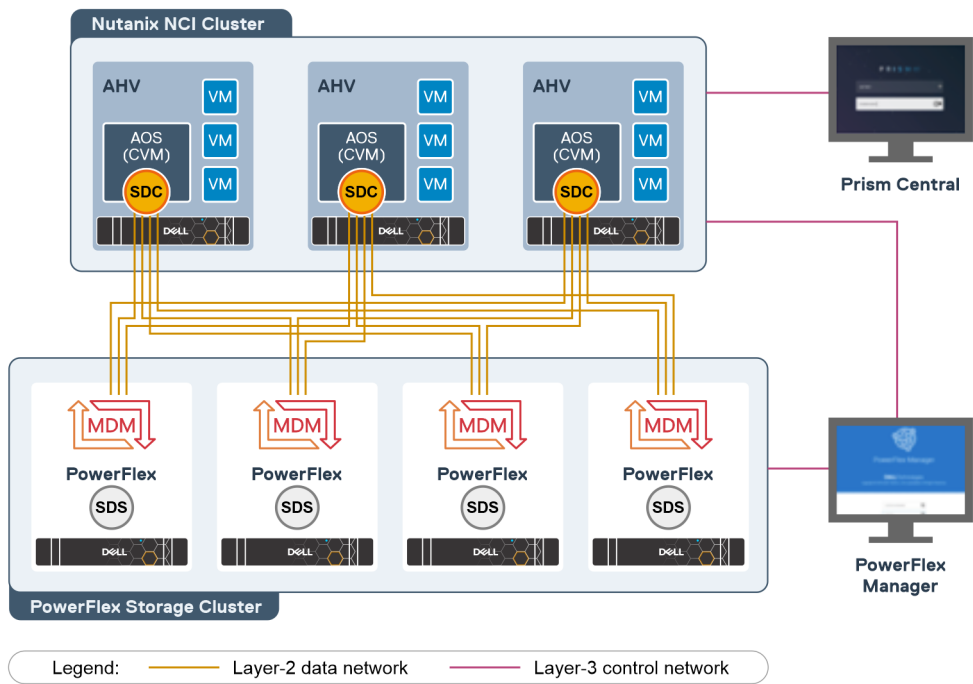
Nutanix Cloud Infrastructure Compute (NCI-C) consists of compute, networking, disaster recovery, and security.

The integration of PowerFlex with Nutanix Cloud Platform enables organizations to scale compute and storage resources independently. Nutanix AHV hypervisor and AOS operate on PowerFlex compute nodes, using storage from an external PowerFlex storage cluster in a two-layer configuration. This setup leverages PowerFlex software-defined storage alongside the Nutanix Cloud Platform infrastructure to optimize resource management. The architecture supports data protection, disaster recovery, and networking while facilitating hybrid cloud operations for scalable workloads.

### Architecture

PowerFlex with Nutanix Cloud Platform is deployed using a two-layer architecture, as shown in the following figure.

- Compute layer: An NCI-C cluster with AOS and AHV running on PowerFlex compute-only nodes.
- Storage layer: A PowerFlex cluster operates in a storage-only model, compatible with any PowerFlex configuration. This cluster provides network-attached storage.



**Figure 9. PowerFlex with Nutanix Cloud Platform architecture**

Unlike the hyperconverged infrastructure (HCI) model, where storage is sourced from locally attached internal drives, the PowerFlex with Nutanix Cloud Platform solution architecture decouples the locally attached internal storage. In this setup, the compute cluster accesses storage over the network from the external PowerFlex storage cluster by the SDC that is installed on every node of the NCI-C cluster.

When you deploy PowerFlex with Nutanix Cloud Platform, you host NCI-C on the PowerFlex compute cluster, while you provision storage from the PowerFlex storage cluster.

## PowerFlex implementation in a VMware ESXi-based system

In the VMware ESXi-based environment, the MDM, and SDS components are installed on a dedicated storage virtual machine, and the SDC is installed directly on the VMware ESXi host.

### Minimum requirements

These are the minimum system requirements for a PowerFlex implementation in a VMware ESXi-based system:

- Three VMware ESXi servers with 240 GB of free capacity per server
- 10 Gbps network

**Table 31. Minimum system requirements**

| Configuration type                                                                                                                                  | Supported VMware ESXi versions                                               | Supported PowerFlex offering                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| VMware ESXi-based hyperconverged nodes with SDC<br><b>NOTE:</b> Compute nodes may also consume storage from VMware ESXi-based hyperconverged nodes. | ESXi 8.0<br><br>Dell Technologies recommends implementing the latest update. | PowerFlex appliance<br>PowerFlex rack                                                                     |
| VMware ESXi compute nodes with SDC connecting to a PowerFlex storage-only system                                                                    | ESXi 8.0<br><br>Dell Technologies recommends implementing the latest update. | PowerFlex appliance<br>PowerFlex rack<br>PowerFlex software<br>PowerFlex custom node<br>VxFlex Ready Node |
| VMware ESXi compute nodes with NVMe over TCP connecting to a PowerFlex storage-only system                                                          | ESXi 8.0<br><br>Dell Technologies recommends implementing the latest update. | PowerFlex appliance<br>PowerFlex rack<br>PowerFlex software<br>PowerFlex custom node<br>VxFlex Ready Node |

**NOTE:** For a full list of supported operating systems, see the [PowerFlex support matrix](#).

### Components

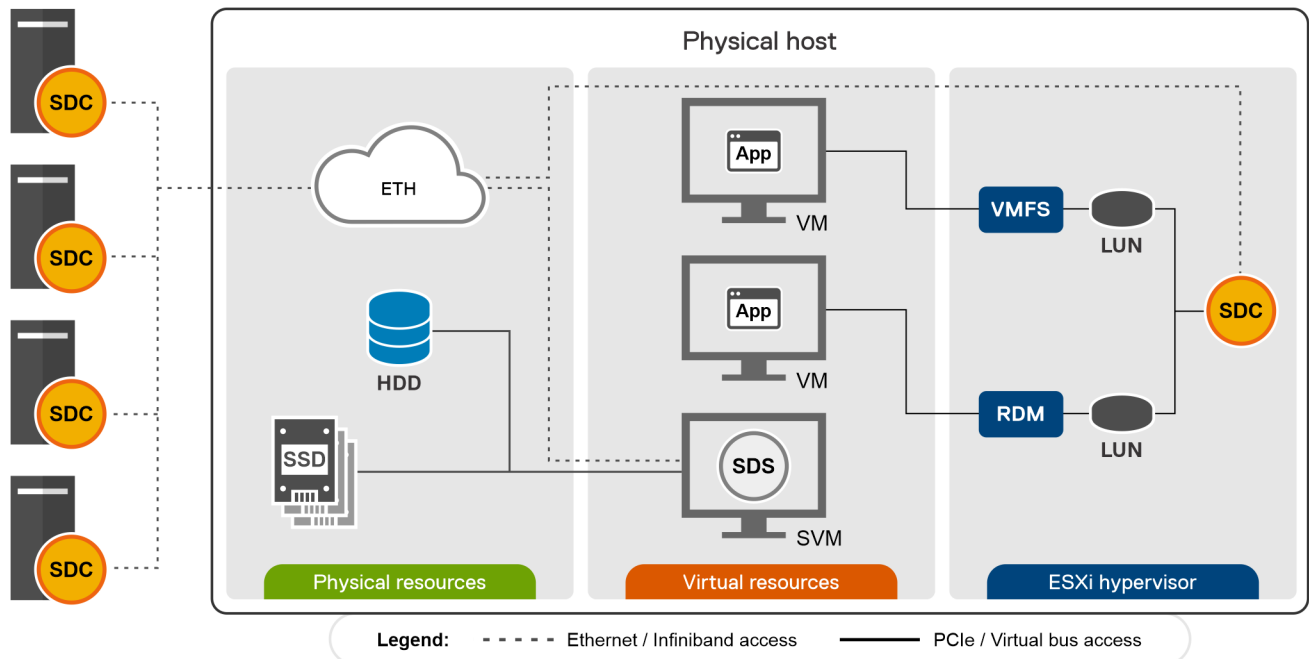
The PowerFlex hyperconverged configuration consists of a dedicated storage virtual machine and these software components:

- Storage virtual machine (storage virtual machine): A Linux-based virtual machine that is dedicated to PowerFlex and is used to host the different PowerFlex software components described here.
- Meta data manager (MDM): Configures and monitors the PowerFlex system. The MDM is configured in a redundant cluster mode. The MDM is installed on the storage virtual machine. The MDM can be configured as a three-node cluster (primary MDM, secondary MDM, and tiebreaker MDM) or as a five-node cluster (primary MDM, two secondary MDMs, and two tiebreaker MDMs) to provide greater resiliency.
- Storage data server (SDS): Manages the capacity of a single server and acts as a backend for data access. The SDS is installed on all servers contributing storage devices to the PowerFlex system. These devices are accessed using SDS. The SDS is installed on the storage virtual machine.
- Storage data replicator (SDR): Handles volumes replication activities between PowerFlex systems through journal management. The SDR is located alongside the SDS.
- Storage data client (SDC): A lightweight device driver that presents PowerFlex volumes as block devices to the application on the server which the SDC is installed. The SDC creates a logical adapter, which is a VMware ESXi kernel construct. The

adapter informs VMware ESXi about the arrival and disappearance of SCSI devices. These LUNs can be formatted with VMFS and then exposed using the VMware ESXi host to the virtual machine or can be used as RDM devices.

**NOTE:** NVMe target (SDT) is not supported in VMware ESXi-based hyperconverged configuration.

This implementation is shown in the figure. The SDC is installed within the VMware ESXi kernel similar to any other VIB.



**Figure 10. PowerFlex implementation in VMware ESXi with SDC in VMkernel**

The LUNs in this example can be formatted with VMFS, and then exposed using the VMware ESXi host to the virtual machine, or the LUNs (volumes) that are NVMe target based can be used as RDM devices. When the LUNs are used as RDM devices, the VMFS layer is omitted.

## Supported features

The following VMware ESXi features are supported by PowerFlex:

- vMotion
- Storage vMotion
- Fault tolerance
- DRS
- Storage DRS
- VAAI (except full copy primitive)

## Device management

Devices can be managed with VMDirectPath I/O or RDM.

- **VMDirectPath I/O** Device management is performed by the storage virtual machine, yielding the best I/O performance. Devices are added to the system after the deployment. On factory-installed VxFlex Ready Node servers, the vSphere plug-in configures DirectPath on the ESXi servers and adds devices to the system. On other servers, you must enable VMDirectPath manually to the ESXi server, then use the plug-in to add devices.
  - When DirectPath is configured, all servers in the system must use DirectPath. For PowerFlex systems, this can be configured with the PowerFlex vSphere plug-in. For other servers, use the vSphere client to configure each ESXi host manually.
  - If the host has multiple controllers, you must configure DirectPath on that host manually, not with the plug-in. After that is done, you can use the plug-in to deploy PowerFlex on that host and to add devices to it.
  - All devices on all servers must not have any other use (not VMDK, RDM, or be part of a datastore).

- Dell PowerEdge 14th generation servers with any NVMe devices cannot be added in a DirectPath-based system. Use RDM-based, instead.
- ESXi boot device requirements:
  - Must be on a separate controller or connected directly to the board.
  - SATADOM and M2 boot devices are supported. These devices allow the creation of a datastore on the system disk, which is required to host the storage virtual machine.
  - USB boot devices are not supported.

 **NOTE:** DirectPath architecture is the recommended best practice for RAID, and SAS Controller managed drives.

## ● RDM

Using RDM mapping, a device is created on the storage virtual machine that points to the hard drive on the ESXi.

You can add RDM devices that are connected through a physical RAID controller. If a local RDM is not connected by a RAID controller, it may not be supported. To ensure the compatibility of these devices, you can add them as VMDK, or you can select **Force RDM on device with unknown RDM support**, as described in the "Advanced settings options" section of the *PowerFlex Deployment Guide*. Enable this option before beginning the deployment.

Before enabling this feature, contact Customer Support.

## ● VMDK

A new datastore is created, with a VMDK, and the VMDK is added to the storage virtual machine. PowerFlex requires thick provisioning, so this process can take a long time.

In almost all cases, RDM is the preferred method to add physical devices. Use the VMDK method only in the following scenarios:

- The physical device does not support RDM.
- The device already has a datastore, and the device is not being completely used. The excess area that is not already being used will be added as a PowerFlex device.

 **NOTE:** To use VMDK, select **Enable VMDK creation**, as described in the "Advanced settings options" section of the *PowerFlex Deployment Guide*.

## System size

If you are deploying a very large PowerFlex system (hundreds of nodes), you can increase the default parallelism limit (default: 100), thus speeding up the deployment. This depends on the processing power of the vCenter.

To increase the parallelism limit, use the plug-in **Advanced settings**, as described in the "Advanced settings options" section of the *PowerFlex Deployment Guide*.

## Pre-deployment considerations

You should take these considerations into account before deploying the system:

- Whether separate networks will be used for data and management (recommended).
- The IP addresses that will be used for the storage virtual machines and VMkernels.
- If flash devices will be added to the SDS .
- If you want to create fault sets. (See the requirements in "Fault sets".)

## Post-deployment considerations

You should take these considerations into account after deploying the system:

- After deployment is complete, set all storage virtual machines to start automatically with the system. Do not set storage virtual machines under the VMware resource-pool feature.
- In a DirectPath environment, after deploying the system, you must add devices to the SDS.

## Other vSphere plug-in features

The plug-in's deployment wizard can be used to register existing PowerFlex systems in vSphere, to add SDS server nodes, or to extend the existing system.

The plug-in can be used to provision PowerFlex nodes.

**NOTE:** When using DirectPath, also known as PCI-passthrough on the storage virtual machine, the following alert appears in vCenter: "Virtual Machine memory Usage." This is a VMware limitation and cannot be resolved. See the [VMware KB](#) for more details.

## VMware ESXi integrated feature support

This section describes VMware ESXi integrated feature support for PowerFlex.

### VMware ESXi vStorage APIs for Array Integration

VMware ESXi vStorage APIs for Array Integration provides hardware acceleration functionality. It allows the host to offload specific virtual machine and storage management operations to compliant storage hardware. With the assistance of the storage hardware, the host performs these operations faster, and consumes less CPU, memory, and storage fabric bandwidth.

VAAI uses these fundamental operations:

- Atomic Test & Set, which is used during creation and locking of files on the VMFS volume.
- Clone Blocks/Full Copy/XCOPY, which is used to copy or migrate data within the same physical array.
- Zero Blocks/Write Same, which is used to zero-out disk regions.
- Thin Provisioning in VMware ESXi 7.x and later hosts, which allows the VMware ESXi host to tell the array when the space previously occupied by a virtual machine (whether it is deleted or migrated to another datastore) can be reclaimed on thin provisioned LUNs.
- Block Delete in VMware ESXi 7.x and later hosts, which allows for space to be reclaimed using the SCSI UNMAP feature.

The PowerFlex supported VAAI features are:

- Atomic Test & Set
- Zero Blocks/Write Same
- Thin Provisioning in VMware ESXi 7.x and later hosts
- Block Delete in VMware ESXi 7.x and later hosts

The following output is an example of typical output::

```
esxcli storage core device vaa1 status get -d
eui.7dbf14034834bbe01bf7e55800000002
eui.7dbf14034834bbe01bf7e55800000002
VAAI Plugin Name:
ATS Status: supported
```

Clone Status: unsupported This means that Clone Block/Full Copy/Xcopy is not supported.

Zero Status: supported This means that write same is supported.

Delete Status: supported This means that UNMAP is supported.

**NOTE:** Thin provisioning is not shown in VAAI output.

### VMware vSphere Storage I/O Control

While VMware Sphere Storage I/O Control (SIOC), Storage I/O Statistics Collection (SIOSC), and Network I/O Control (NIOC) are useful for vSan environments, their implementation may cause significant issues in a PowerFlex environment. Therefore, use of these options is not supported.

PowerFlex provides built-in capabilities to limit network bandwidth and IOPS limits for each volume for each storage data client.

## Supporting VMware ESXi

PowerFlex Manager supports VMware ESXi 8.0 and VMware vCenter 8.0. VMware vCenter 8.0 supports the deployment of VMware ESXi-based hyperconverged infrastructure or managed compute nodes with VMware ESXi 8.0 and service operation. VMware ESXi 8.0 support is equivalent to the current support for VMware ESXi 7.x.

### VMware ESXi 8.0

Upgrade from VMware ESXi 7.x to VMware ESXi 8.x by performing a non-disruptive update operation on existing deployments. Install VMware ESXi 8.x on new deployments of PowerFlex.

### Upload complete VMware ESXi software bundles

Upload the entire VMware ESXi software bundle in a ZIP format, from a local drive or a network location. On completion of the node discovery activity, the BIOS configuration is implemented using iDRAC.

### VMware ESXi upgrade

PowerFlex Manager supports enabling the VMware ESXi upgrade and allows upload of external VMware ESXi upgrade images.

The following upgrades can be done using PowerFlex Manager for appliance and rack systems that are on 4.x:

- VMware ESXi 8.x to 8.x upgrades
- VMware ESXi 7.x to 8.x upgrades
- VMware ESXi 7.x to 7.x upgrades

## Consumption models

PowerFlex is offered in the following consumption models: PowerFlex appliance, PowerFlex rack, PowerFlex software, PowerFlex custom node, and VxFlex Ready Node.

## PowerFlex appliance

PowerFlex appliance is a scalable system with flexible form factors that comes pre-configured and validated for fast, easy deployment.

**Table 32. PowerFlex appliance components**

| Component                                                                                                       | PowerFlex appliance Intelligent Catalog                                               |
|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| PowerFlex software                                                                                              | Included                                                                              |
| Operating system and drivers                                                                                    | Included                                                                              |
| System firmware <ul style="list-style-type: none"><li>• BIOS</li><li>• Cards</li><li>• Drives</li></ul>         | Included                                                                              |
| Infrastructure firmware <ul style="list-style-type: none"><li>• Cabinet / PDU</li><li>• IPI Appliance</li></ul> | N/A                                                                                   |
| CloudLink                                                                                                       | Included                                                                              |
| VMware NSX                                                                                                      | Customer-managed                                                                      |
| Network switch firmware                                                                                         | Full network automation is included<br><br>Partial network automation: Support Matrix |



## PowerFlex rack

PowerFlex rack is a rack-scale system that is manufactured, managed, supported, and sustained as one system for single-end-to-end lifecycle support.

**Table 33. PowerFlex rack components**

| Component                                                                                                     | PowerFlex rack Release Certificate Matrix |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| PowerFlex software                                                                                            | Included                                  |
| Operating system and drivers                                                                                  | Included                                  |
| System firmware <ul style="list-style-type: none"><li>• BIOS</li><li>• Cards</li><li>• Drives</li></ul>       | Included                                  |
| Infrastructure firmware <ul style="list-style-type: none"><li>• Cabinet/PDU</li><li>• IPI Appliance</li></ul> | Included                                  |
| CloudLink                                                                                                     | Included                                  |
| VMware NSX                                                                                                    | Customer-managed                          |
| Network switch firmware                                                                                       | Included                                  |

## PowerFlex software

The following table describes the components in the PowerFlex software offering:

**Table 34. PowerFlex software components**

| Component                                                                                                     | Software management catalog                                                       |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| PowerFlex software                                                                                            | Included                                                                          |
| Operating system                                                                                              | Not included. See the <i>PowerFlex Support Matrix</i> for your PowerFlex version. |
| System firmware <ul style="list-style-type: none"><li>• BIOS</li><li>• Cards</li><li>• Drives</li></ul>       | Customer managed                                                                  |
| Drivers                                                                                                       | Customer managed                                                                  |
| Infrastructure firmware <ul style="list-style-type: none"><li>• Cabinet/PDU</li><li>• IPI Appliance</li></ul> | N/A                                                                               |
| Dell CloudLink                                                                                                | Support Matrix                                                                    |
| VMware NSX                                                                                                    | Customer managed                                                                  |
| Network switch firmware and operating system                                                                  | Customer managed                                                                  |

## PowerFlex custom node and VxFlex Ready Node

PowerFlex custom node is a do-it-yourself networking and management solution in which PowerFlex is installed on Dell PowerEdge servers that have been optimized for PowerFlex systems.

**Table 35. PowerFlex custom node and VxFlex Ready Node components**

| Component                                                                                                           | Software management catalog             |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| PowerFlex custom node and VxFlex Ready Node                                                                         | Included                                |
| Operating system                                                                                                    | See the <i>PowerFlex Support Matrix</i> |
| System firmware and drivers <ul style="list-style-type: none"><li>• BIOS</li><li>• Cards</li><li>• Drives</li></ul> | See the <i>PowerFlex Support Matrix</i> |
| Infrastructure firmware <ul style="list-style-type: none"><li>• Cabinet/PDU</li><li>• IPI Appliance</li></ul>       | N/A                                     |
| CloudLink                                                                                                           | See the <i>PowerFlex Support Matrix</i> |
| VMware NSX                                                                                                          | Customer-managed                        |
| Network switch firmware                                                                                             | Customer-managed                        |

## Additional functions

This section lists additional functions of PowerFlex.

### Physical layer

To implement the physical layer, perform the following steps:

1. Install the MDM component on the MDM nodes in one of the following configurations:
  - Three-node redundant cluster (one primary MDM, one secondary MDM, and one tiebreaker).
  - Five-node redundant cluster (one primary MDM, two secondary MDMs, and two tiebreakers).
  - Single node (one primary MDM).

**NOTE:** It is not recommended to use single mode in production systems, except in temporary situations. The MDM contains all the metadata that is required for system operation. Single Mode has no protection, and exposes the system to a single point of failure.

MDMs do not require dedicated nodes. They can be installed on nodes hosting other PowerFlex components.

2. Install the SDS component on all nodes that contributes some, or all, of their physical storage. Typically, each SDS is physically on a separate server.
  - Divide the SDS nodes into protection domains. Each SDS can be a member of only one protection domain.
  - Per protection domain, divide the physical storage units into storage pools, and optionally, into fault sets.
3. Install the SDC component on all nodes on which the application accesses the data exposed by the PowerFlex volumes.
  - In the following figure, the gray boxes represent a storage node which hosts the SDS, SDC, and MDM where applicable. In the cases where the gray boxes host the SDS, SDC, and MDM it is a three-node MDM cluster.



**Figure 11. Physical layout example—three-node MDM cluster**

Communication is done over the existing LAN using standard TCP over IP. The MDM and SDS nodes can be assigned to up to eight IP addresses, enabling wider bandwidth and better I/O performance and redundancy.

## SAN virtualization layer

After you install and configure the storage data client (SDC) components, expose the virtual SAN devices to the servers on which the SDCs are installed.

The MDM cluster manages the entire system. It aggregates the entire storage that is exposed to it by all the SDSs to generate a virtual layer - virtual SAN storage. Volumes can now be defined over the storage pools and can be exposed to the applications as a local storage device using the SDCs.

To expose the virtual SAN devices to your servers (the ones on which you installed and configured SDCs), perform the following:

1. Define volumes. Each volume that is defined over a storage pool is evenly distributed over all members using a RAID protection scheme. By having all SDS members of the storage pool participate, PowerFlex ensures:
  - Highest and most stable and consistent performance possible
  - Rapid recovery and redistribution of data
  - Massive IOPS and throughput

You can define volumes as follows:

- Thick

Capacity is allocated immediately, even if not actually used. This can cause capacity to be allocated, but never used, leading to wasted capacity.

Thick capacity provisioning is limited to available capacity.

- Thin

Capacity is “on reserve,” but not allocated until actually used. This policy enables more flexibility in provisioning.

Whereas thick capacity is limited to available capacity, thin capacity provisioning can be oversubscribed, as follows:

Maximum thin capacity provisioning = 5 \* (total raw capacity - used capacity)

When capacity usage reaches the level where it may cause I/O errors, alerts are generated. At certain higher capacity levels, volumes (even thin volumes) can no longer be created.

Example:

In a system with 3 SDSs, each with 10 TB, there are 30 TB of storage.

In the system, there is already a thick-provisioned volume that takes up 15 TB of the total raw capacity (created by adding a 7.5 TB volume).

MDM will allow a total of 300 TB total raw capacity to be provisioned, and since 15 TB are already allocated, you can add a thin-provisioned volume of 285 TB total raw capacity (by adding a 142.5 TB volume) or a thick-provisioned volume of 15 TB total raw capacity.

 **NOTE:** This example uses 10x fine granularity over provisioning plus compression support.

2. Map volumes. Designate which SDCs can access the given volumes. This gives rise to the following:

- Access control per volume exposed
- Shared nothing or shared everything volumes

Once an SDC is mapped to a volume, it immediately gets access to the volume and exposes it locally to the applications as a standard block device. These block devices appear as `/dev/sciniX` where `X` is a letter, starting from “a.”

For example:

- `/dev/scinia`
- `/dev/scinib`

- The maximum amount of partitions for the scini disk is 15.

3. The maximum amount of volumes that can be mapped to an SDC is listed in the “Supported capabilities and system limits” table.

 **NOTE:** SDC mapping is similar to LUN mapping, in the sense that it only allows volume access to clients that were explicitly mapped to the volume.

## Audit logging

PowerFlex enables the redirection of the PowerFlex events and Ingress audit messages to syslog.

These events and audit messages are forwarded to an external Security Information and Event Manager (SIEM). You must perform the following configuration steps to enable this feature in PowerFlex Manager:

- Change Ingress settings to emit audit messages.
- Define a notification policy to forward Ingress audit messages in the PowerFlex system to an SIEM.
- Define a notification policy to forward events in the PowerFlex system to an SIEM.

For more information about the configuration steps, see the *Dell PowerFlex and PowerFlex Manager Administration Guide*.

## Supported events

Audit logging supports the following event categories that occur in PowerFlex Manager and integrated components and systems:

- User authentication and authorization
- Changes to system configuration and security policies
- Security events

## Maintenance

Maintenance of PowerFlex is primarily limited to configuration changes of the physical and virtual layers. It requires minimal user attention. When maintenance or planned restart of an SDS is required, the maintenance mode feature can be used to streamline system operation.

### Maintain the physical layer

Adding or removing hardware units and configuring them into PowerFlex can result from scale-out, hardware failure, operating system updating, or upgrade and hardware expansion.

In the physical layer, maintenance is limited to adding and removing hardware units and configuring them into PowerFlex. These operations are usually a result of:

- Scaling out when there is a need for additional capacity. This usually results in adding more storage media to the existing servers, or adding additional servers.
- Hardware failure. In cases where there is a hardware (storage media or server) failure and it must be replaced.

In all the above cases, the operation requires adding or removing storage capacity from the system. In some cases, it may include adding or removing an entire server, and its associated storage media, from the configuration. As far as PowerFlex is concerned, all these activities translate to SDS reconfigurations.

If the removed server is an SDC node, or the server to be added requires exposing storage locally, SDC reconfiguration will happen as well.

- Adding or removing storage media. Add or remove the media from the SDS with which it is associated. PowerFlex will redistribute the data accordingly and seamlessly.
- Operating system updating, or upgrade or hardware expansions. Graceful shutdown or reboot operations with return to fully operation state flow.

### Instant maintenance mode

Using instant maintenance mode, you can restart a server hosting an SDS without requiring a shutdown or interrupting application I/Os. Storing all writes created during maintenance to a dedicated fault set prevents data loss if there is a single failure.

Instant maintenance mode enables you to restart a server that hosts an SDS, with minimal impact on PowerFlex, thus bypassing the disruption and effort caused by disorderly shutdown, protection domain shutdown, and orderly shutdown.

Whereas PowerFlex always uses two copies of user data, invoking maintenance mode introduces an additional copy that stores all writes created during maintenance to an SDS or fault set (created during maintenance) in both a primary location and a new location. This copy prevents data loss if a single failure occurs.

When the SDS or fault set is returned from maintenance mode, only the new writes are required to be resynchronized, thus minimizing data transfer during and after the update.

Instant maintenance mode does not interrupt application I/Os; it can be run on any amount of members of a fault set; and it can run in parallel on different protection domains. While an SDS is in maintenance mode, most PowerFlex operations (like adding a volume) cannot be performed in the fault set, protection domain, or storage pool in which the SDS and its devices reside.

To invoke maintenance mode, the following conditions are required:

- Only one fault unit (or stand-alone SDS) can be in maintenance mode at any given time.
- No other SDSs can be in a degraded or failed state (force override can be used).

- There must be adequate space on other SDSs for the additional backup (force override can be used).

**NOTE:** Use of force override options when entering maintenance mode can lead to data unavailability while maintenance mode is activated.

While an SDS is in maintenance mode, it can be shut down with no danger to data.

## Protected maintenance mode

Protected maintenance mode is another type of maintenance mode that ensures no data loss during down time.

Protected maintenance mode is another type of maintenance state where a third copy is created. If there should be a node failure during instant maintenance mode, then there is no SDS up to serve I/Os for the data. With protected maintenance mode, a third copy is created before maintenance mode, which ensures that if there is a node failure where the second copy of SDS is required, there is still a full backup of the SDS.

### Auto abort

Protected maintenance mode consumes a significant amount of capacity. In some cases, this capacity may be required for more important tasks such as user I/O or rebuild.

The purpose of this capability is to cancel protected maintenance mode when approaching the end of capacity, to avoid impacting I/O or rebuild. For example, if a single node is entering protected maintenance mode and a third copy is being created, then a second node fails. Auto abort will confirm that the third copy space is made available to perform a rebuild.

## Maintain the virtualization layer

This section lists the operations that can be performed on volumes exposed by the PowerFlex virtual SAN.

The following operations may be performed on volumes that are exposed by the PowerFlex virtual SAN:

- Add or remove a volume:  
Create or delete a volume in the system.
- Increase volume size:  
Add capacity to a given volume, as needed. The change in volume size occurs seamlessly without interrupting I/O.
- Map and unmap volumes to an SDC:  
This enables or disables access to a volume by an SDC, and thus by an application residing on the same node.
- MDM operations:
  - Move the MDM role to another role
  - Add MDM VIP
  - Upgrade of MDM, SDS, SDC and LIA

## Decommissioned features

Read Flash Cache (RFCache) has been removed from the Powerflex 4.0 product offering.

When upgrading a system to 4.0 that has RFCache, you must remove the RFCache configuration and files before proceeding with the upgrade.

## PowerFlex management

PowerFlex management covers all product functionalities, including block storage, hardware management, network management, reporting, and alerts.

- The management can be accessed through a Web UI or REST API.
- PowerFlex Manager handles user management for all systems, including:

- PowerFlex local users
- LDAP/AD users
- PowerFlex Manager supports system security, as follows:
  - PowerFlex management platform exposes an HTTPS interface for Web UIs and REST APIs.
  - Users must have an account that is defined in PowerFlex.
  - Upon login, users receive an access token.
  - PowerFlex Manager uses OpenID Connect (OIDC).
  - PowerFlex management platform communication with PowerFlex block storage systems is secured using mTLS.
- PowerFlex management platform is a cluster of three or more self-managed Kubernetes nodes, providing resiliency and high availability.
  - PowerFlex management platform run time optimized to 60 minutes.
- The following CLI tools are available:
  - A CLI that provides full control over the block storage system.
  - A CLI with expanded support for management, reporting, and alerting.
- Enhanced security strength by using SSH keys and non-root users through sudo in management aspects.

For deployment and post-deployment procedures, see the applicable versions of the *Dell PowerFlex and PowerFlex Manager Install and Upgrade Guide* and *Dell PowerFlex and PowerFlex Manager Administration Guide* for configuration guidelines.

- Webhooks support enables sending alerts to the webhooks servers designed for software platforms such as BigPanda.
- Multi-subnet and Multi-VLAN for PowerFlex appliance and PowerFlex rack allows defining multiple subnets and VLANs in a single network.

The most common implementation example is with a management or vMotion network that spans multiple racks. Each rack is its own subnet and VLAN.

 **NOTE:** Supported on all networks except for the following: PowerFlex data, vSAN, and NSX external networks.

- Higher flexibility of fault sets defines a subset of nodes in a rack as part of a fault set during deployment and expansion operations.

For example, a rack with 20 PowerFlex nodes can have four fault sets, one for every five nodes, for easier maintenance support. Another option is using fault sets in a full rack configuration, such as one fault set per rack.

- PowerFlex management cluster self-awareness on PowerFlex appliance and PowerFlex rack allows self-awareness of its underlying controller system.
  - Upgrade from end to end and receive alerts from issues in the PowerFlex management controller infrastructure.
- Modular (single) component upgrade changes the package in the PowerFlex rack and PowerFlex appliance catalogs to newer or older versions.

This upgrade depends on customer cases that want to deviate from the Release Certification Matrix or Intelligent Catalog.

 **NOTE:** By default this feature is turned off and can only be enabled through an RPQ process. If applicable, contact your account team for details.

## PowerFlex Manager

PowerFlex Manager is the unified management user interface for all PowerFlex systems. It allows you to deploy and manage new and existing PowerFlex systems, and also includes life cycle and resource management capabilities.

PowerFlex Manager provides IT operations management for PowerFlex systems. It increases efficiency by reducing time-consuming manual tasks that are required to manage system operations. PowerFlex Manager can also be used to deploy and manage new and existing PowerFlex appliance and PowerFlex rack systems.

PowerFlex Manager includes the following tabs:

- A dashboard that enables you to see a summary of performance and PowerFlex appliance and PowerFlex rack inventory data.
- A block tab that provides a user interface for managing PowerFlex block storage.
- A protection tab that provides a user interface for performing PowerFlex storage protection operations.
- A lifecycle tab that allows you to create and clone templates for PowerFlex appliance and PowerFlex rack and deploy them as resource groups (previously known as services).
- A resources tab that allows you to manage all resource types within the system.

- A monitoring tab that allows you to track events, alerts, and jobs running.
- A settings tab that enables you to configure system settings for PowerFlex Manager.

PowerFlex Manager is part of all PowerFlex consumption models.

PowerFlex Manager is designed as microservices, specialized modules working together and based on the Rancher Kubernetes Engine.

## Management capabilities

PowerFlex Manager provides the unified interface and full lifecycle mode services. Role-based and scope-based access control enables common PowerAPIs to improve ecosystem and customer integration of the PowerFlex storage solution.

PowerFlex provides one tool for all consumption models.

- PowerFlex Manager (UX, API, CLI)
  - Unified dashboard
  - REST API for PowerFlex rack, PowerFlex appliance, and PowerFlex software
  - PowerFlex Block and PowerFlex file services
  - SDC and NVMe over TCP hosts
- Unified services:
  - Security
  - Single sign-on (SSO) and role-based access control (RBAC)
  - Deployment and LCM
  - Events, alerts, and reporting
  - Logging and metrics collection
  - SNMP, Secure Remote Services, SMTP, CloudIQ
- Unified RBAC and user management
  - No separate users or LDAP configuration
  - Single set of roles
  - Meta data manager (data-path) does not maintain RBAC for PowerFlex 4.0 and later.
- Single WebUI
- Single REST endpoint for all APIs: Legacy and modern, ISG-standard PowerAPI

 **NOTE:** For login information, see the PowerFlex API documentation on the [Dell Technologies Developer](#) portal.

- Events (log) and alerts (state) including meta data manager (MDM) events
- Resilient management stack

## Authenticating with a single sign-on and a common identity provider

PowerFlex Manager supports authentication for single sign-on (SSO) using a common identity provider (IdP).

This support relies on a multicloud solution based on Red Hat OpenShift using PowerFlex. With this feature, PowerFlex introduces an OpenID Connect (OIDC) IdP provider type as an alternative to SAML, enabling a seamless user experience leveraging SSO to PowerFlex.

## PowerFlex Manager and VMware vCenter automation and integration

PowerFlex Manager has extensive orchestration capabilities with VMware vCenter and ESXi.

For orchestration capabilities, see the following for the supported feature set and dependencies:

 **NOTE:** You must supply the relevant license to support the relevant feature set.

**Table 36. PowerFlex Manager and VMware vCenter automation and integration**

| Features                         | Dependencies                     |
|----------------------------------|----------------------------------|
| Deploy and configure VMware ESXi | iDRAC access to a PowerFlex node |



**Table 36. PowerFlex Manager and VMware vCenter automation and integration (continued)**

| Features                                                                                                | Dependencies                                                          |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Deploy and configure VMware ESXi with persistent memory                                                 | iDRAC access to a PowerFlex node with persistent memory (SDPM/NVDIMM) |
| Create VMware ESXi cluster                                                                              | VMware vCenter access                                                 |
| Set up VMware ESXi HA and DRS                                                                           | VMware vCenter access with vCLS enabled                               |
| Create and configure VMware vCenter based vSphere distributed switch                                    | VMware vCenter access                                                 |
| Create datastore and deploy VMs                                                                         | Access to VMware vCenter and VMware ESXi                              |
| Service and lifecycle VMware ESXi compute-only nodes active workloads                                   | Access to VMware vCenter and VMware ESXi                              |
| Service and lifecycle VMware ESXi hyperconverged-only nodes with persistent memory and active workloads | Access to VMware vCenter and VMware ESXi                              |
| Enable or disable SSH to VMware ESXi upon demand                                                        | Access to VMware ESXi                                                 |

## Supporting non-bonded networking configurations

PowerFlex Manager supports non-bonded networking options with resource group templates.

Configure a storage-only template to use a non-bonded networking configuration where data networks are not bonded. Configure a compute-only template to work with a non-bonded storage-only deployment. To enable non-bonded networking, PowerFlex Manager allows the same PowerFlex management network to be on two ports on the same interface.

Set up two different types of non-bonded configurations on a resource group template:

- Four data network configuration
  - The PowerFlex management network is selected on all ports in interface1 and interface2. The management network is set on Bond0 for this configuration. This configuration reserves only one PowerFlex management IP address.
  - Data 1 is on port 1 of the first interface (assigned to Eth 0 with no bond).
  - Data 2 is on port 2 of the first interface (assigned to Eth 1 with no bond).
  - Data 3 is on port 3 of the second interface (assigned to Eth 2 with no bond).
  - Data 4 is on port 4 of the second interface (assigned to Eth 4 with no bond).
- Two data network configuration
  - The PowerFlex management network is selected on port 1 in interface1 and interface2. The management network is set on Bond0 for this configuration. This configuration reserves only one PowerFlex management IP address.
  - Data 1 is on port 1 of the second interface (assigned to Eth 0 with no bond).
  - Data 2 is on port 2 of the second interface (assigned to Eth 1 with no bond).

For a non-bonded storage-only template, you must select **Trunk Port** for **Switch Port Configuration** setting, not **Port Channel (LACP enabled)**. For the **Teaming and Bonding Configuration** setting, you typically select **Mode 1** for most deployments.

PowerFlex Manager provides eight different sample templates that support non-bonded network configurations. Four of the templates are for storage-only and four are for compute-only deployments. Six of the templates are new and two are modifications to existing templates offered in previous releases:

- Storage-25gb-4Data-Trunked (replaces existing Storage-Non-Bonding template)
- Storage-100gb-4Data-Trunked
- Storage-25gb-2Host-Data-Trunked
- Storage-100gb-2Host-Data-Trunked
- Compute-Linux-25gb-2Data-Trunked (replaces existing Compute-Linux template)
- Compute-Linux-25gb-4Data-Trunked
- Compute-Linux-100gb-4Data-Trunked
- Compute-ESXi-25gb-2Data-Trunked

## High availability of the PowerFlex Manager user interface

PowerFlex Manager is able to recover from most failures in a short amount of time, enabling continuous management and operation of the PowerFlex system.

When a node is not available, a banner is displayed at the top of the screen indicating that the system is undergoing maintenance and you must wait up to 10 minutes for recovery. Once the node returns to normal status, the banner disappears. Whenever the user interface is not able to retrieve data from the backend, PowerFlex Manager displays a message indicating that no data is available due to maintenance. The block, alerts, and events services run replicas of the data to allow the relevant parts of the user interface to remain accessible when a node or pod fails.

## Serviceability

PowerFlex Manager provides support through integration with the secure connect gateway ensuring better alignment with Dell Technologies Services initiatives to enhance the user experience.

Registering your device with SupportAssist through the secure connect gateway, enables you to do the following:

- Collect system state information and telemetry.
- Automate issue detection and support case creation.
- Resolve common issues remotely.
- Provide critical information to Dell Technologies remote support to resolve complex issues.

## Deployment options for each management offering

PowerFlex management platform is an integrated part of the PowerFlex system. PowerFlex management platform can be set up in multiple deployment options, regardless of the consumption model.

Depending on which consumption model you choose, you have different management options within the solution stack. PowerFlex Manager manages some aspects of the solution while other elements are customer-managed.

The supported deployment options available are as follows:

- Co-resident: Production nodes and storage hosts management instances of Kubernetes and containers.
- Bring your own hypervisor: Your hypervisor and shared storage hosts management VMs with Kubernetes.
- Single management node: A few disks in RAID 5 for shared storage hosts management VMs with Kubernetes.
- This option uses three or more nodes that are configured to create a Management VMware ESXi cluster running a management PowerFlex cluster which provides the shared storage datastore for management operations.

**Table 37. Deployment option per offering**

| Supported offering                                | Co-resident | Bring your own hypervisor                                                                    | Single management node | Three or more management nodes |
|---------------------------------------------------|-------------|----------------------------------------------------------------------------------------------|------------------------|--------------------------------|
| <b>PowerFlex software</b>                         | Yes         | Yes                                                                                          | No                     | No                             |
| <b>PowerFlex custom node or VxFlex Ready Node</b> | Yes         | Yes                                                                                          | Yes                    | No                             |
| <b>PowerFlex appliance</b>                        | No          | Yes<br><br>Not available by default. Contact Dell Technologies Support for more information. | Yes                    | Yes                            |
| <b>PowerFlex rack</b>                             | No          | Yes<br><br>Not available by default. Contact Dell Technologies Support for more information. | No                     | Yes                            |

**NOTE:** QCOW2 images are available on the Dell Technologies support site and can be downloaded by the customer for a bring your own Kernel-based Virtual Machine environment with a basic maintenance of virtual machine health check, shut down, and start up procedures.

The QCOW2 images enable the deployment of PowerFlex management platform, CloudLink, and jump server (exclusively for new appliances).

## Dedicated management cluster model

The PowerFlex management platform can be deployed on dedicated management nodes or clusters running outside of PowerFlex storage-only nodes, PowerFlex compute-only nodes, or PowerFlex hyperconverged nodes.

When you deploy a dedicated management cluster, the PowerFlex management platform manages the complete solution stack.

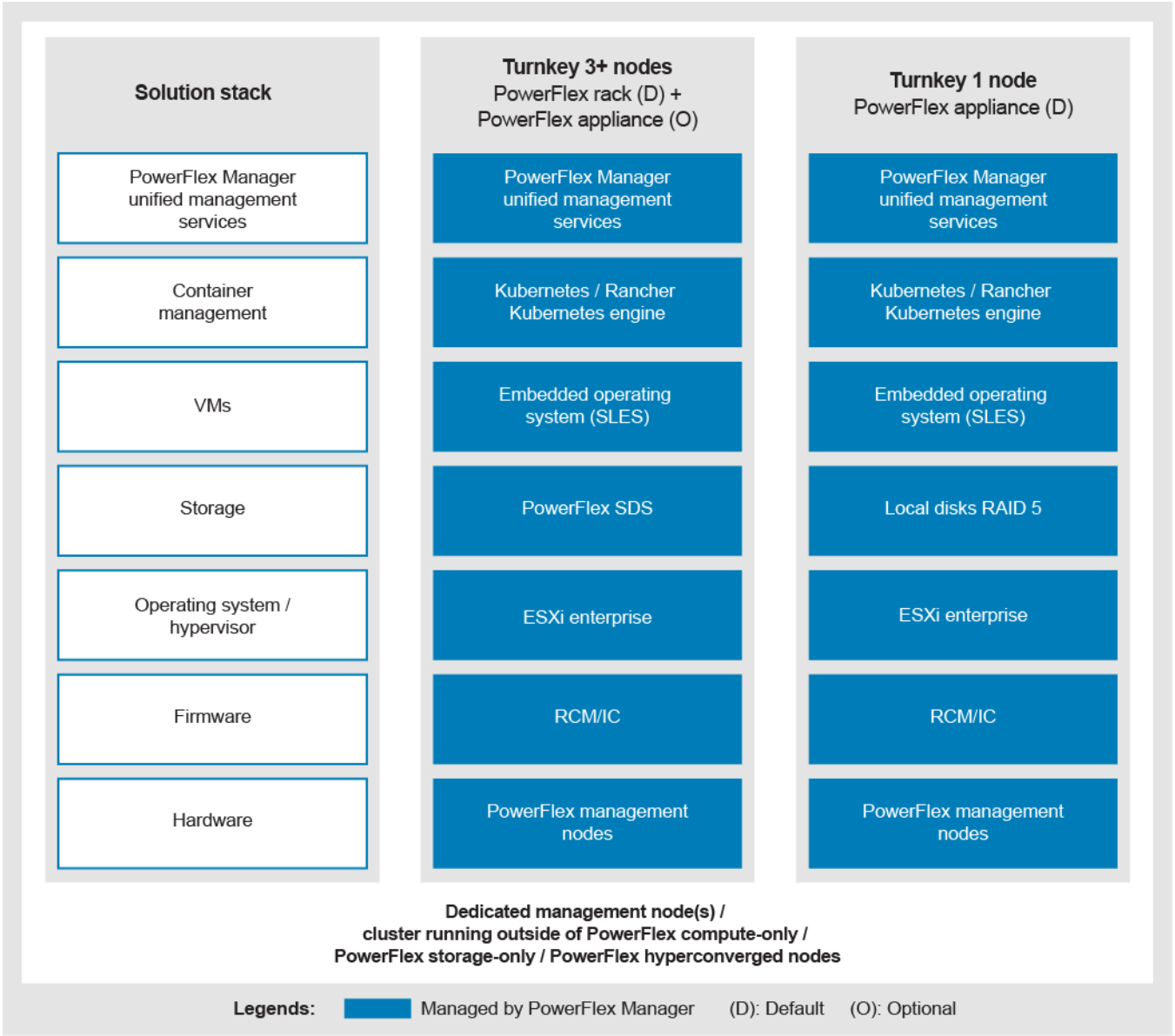


Figure 12. Dedicated management cluster model

# Co-resident software management model

The PowerFlex management platform supports a co-residency deployment model in which the PowerFlex management platform is deployed on the same machines as PowerFlex storage-only nodes.

In the co-resident deployment model, the PowerFlex management platform is deployed on the boot device of three PowerFlex storage nodes so that the V and PowerFlex storage co-reside on the same node. For more information, see the *Dell PowerFlex and PowerFlex Manager Install and Upgrade Guide*.

In this management option, the PowerFlex management platform manages only the containers and PowerFlex, as illustrated in the following diagram:

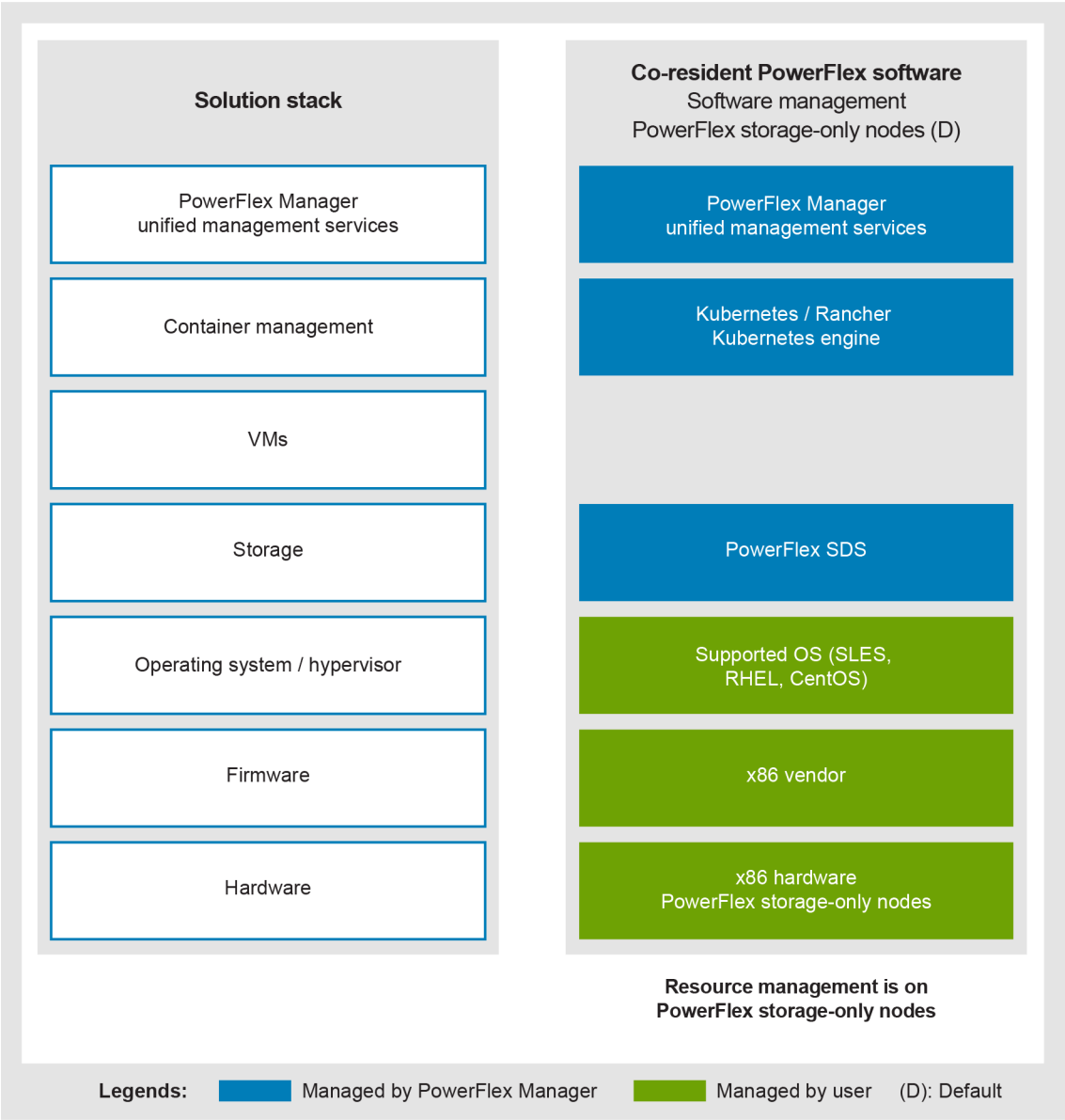


Figure 13. Co-resident software management model

# Non-co-resident bring your own management model

The PowerFlex management platform supports non-co-residency, where it can be deployed on a supported operating system of your choice. In this management option, the PowerFlex management platform manages only the containers and PowerFlex.

**NOTE:** For non-co-resident, bring your own Kernel-based Virtual Machine models, customers can download QCOW2 images from the Dell Technologies support site with a basic maintenance of virtual machine health check, shut down, and start up procedures.

The QCOW2 images enable the deployment of PowerFlex management platform, CloudLink, and jump server (exclusively for new appliances).

The following image explains the solution stack, depending on the operating system:



Figure 14. Non-co-residency bring your own management model

## Configure direct attached storage

When adding direct attached storage to SDS, it is recommended to configure the raw devices as stand-alone devices. Upon adding a device to an SDS PowerFlex verifies that the device is clear and returns an error if it is otherwise.

PowerFlex works with any free capacity—internal or direct-attached devices, either magnetic hard disk drives or flash-based devices such as solid state drive (SSD) and PCIe cards. Although PowerFlex can work with any device topology, it is recommended to configure the raw devices as stand-alone devices.

Device data is erased when devices are added to SDS. When adding a device to an SDS, PowerFlex checks that the device is clear before adding it. An error is returned, per device, if it is found not to be clear. You can override this check by using the force device takeover option.

The following devices are considered to be not "clear," and thus cannot be added to SDS:

- Linux - A complete device with either a file system or partition, or a partitioned device with a file system.
  - Multipath devices cannot be added as SDS devices.
  - Devices in an LVM group cannot be added to an SDS.
- VMware ESXi - Same as above, depending on the operating system of the SVM where the SDS is installed.

Limitations:

- SAN devices will not be prevented from being added.
- Devices in an LVM group cannot be added to an SDS.
- Within the database devices, only Oracle ASM devices can be detected and blocked.

 **NOTE:**

If the server has a RAID controller, PowerFlex prefers to use the caching abilities of the controller for better performance, but is better used when all devices are configured as stand-alone (that is setting each of the devices to RAID 0 separately). Dell Technologies recommends enabling RAID controller caching for hard disk drive devices. Caching is not needed for SSD devices.

 **NOTE:**

For hard disk drives: It is recommended to use RAID controller caching when available as follows:

- READ/WRITE: If the cache is battery-backed
- READ ONLY: If the cache is NOT battery-backed

For SSDs (such as Flash): The setting should be write-through.